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DM_Ch2_Getting_to_Know_Your_Data.pdf
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each topic make it verbose with example and practive proble. why it is used and scenario each page each topic this way make me understand

ChatGPT gpt-6

I'll turn your **Data Mining – Chapter 2: Getting to Know Your Data** lecture into a detailed, beginner-friendly learning guide, following the exact order of your uploaded PDF.

I won't just explain what a term means. I want you to understand **what it does, why we need it, when to use it, how to calculate it, and how it appears in real-world data mining problems.**

For every topic, I'll follow this structure:

1. **Concept:** Explanation from the lecture in simple language.
2. **Why is it used?** The actual purpose in data mining.
3. **Real-world scenario:** Where you would encounter it.
4. **Worked example:** Step-by-step calculation or demonstration.
5. **Practice problem:** A question for you to solve.
6. **Solution:** Detailed explanation so you can verify your understanding.
7. **Exam tips:** Important formulas, distinctions and common mistakes.

I'll preserve the lecture's terminology, examples and formulas, while adding explanations and practical scenarios to make the material easier to understand.

Data Mining – Chapter 2: Getting to Know Your Data

Part 1: Understanding data and attribute types

Page 1–2: Introduction – Why do we need to know our data?

1. What is the purpose of this chapter?

Imagine you work at a brokerage company and have a dataset containing 100,000 stock transactions.

Each transaction contains:

- Stock name
- Transaction price
- Trading volume
- Transaction date
- Broker ID
- Whether the transaction is suspicious
- Market category

Now suppose your manager asks you to build a machine learning model that identifies suspicious transactions.

Would you immediately apply a clustering algorithm?

No. First, you need to understand your data.

Why?

Because different types of data require different statistical methods, visualizations and similarity calculations.

For example:

- You can calculate the average transaction price.
- You cannot meaningfully calculate the average stock name.
- You can calculate the difference between two transaction prices.
- You cannot calculate a meaningful numerical difference between two broker IDs.

This is the fundamental idea behind this chapter.

2. Why study data before applying algorithms?

Every data mining algorithm makes assumptions.

Consider three examples:

Algorithm	Assumption	What can go wrong?
K-Means	Mean and numerical distance are meaningful	Categorical data encoded as numbers may produce meaningless clusters

Algorithm	Assumption	What can go wrong?
K-Nearest Neighbors	Distance represents actual similarity	Different feature scales can distort distance
Linear Regression	Relationships between variables can be modelled numerically	Outliers can significantly affect the fitted model

Example:

Suppose you have two customer attributes:

Customer	Age	Annual income
A	25	300,000
B	30	500,000

If you calculate Euclidean distance directly:

$$d(A, B) = \sqrt{(25 - 30)^2 + (300000 - 500000)^2}$$

$$d(A, B) \approx 200000$$

The income difference dominates the calculation. The age difference becomes almost irrelevant.

This can lead to incorrect customer similarity measurements.

The solution is to understand and normalize the attributes before calculating distance.

3. Four questions to ask before data mining

Question 1: What kinds of attributes do we have?

Are they numerical, categorical, binary or ordinal?

This determines which calculations are appropriate.

Question 2: What do the values look like?

You need to understand:

- Centre: Where do most values lie?
- Spread: How widely are values distributed?

- Skewness: Is the distribution balanced or asymmetric?
- Outliers: Are there unusually large or small values?

Question 3: Can we visualize the structure?

Graphs help you identify patterns that may not be obvious from tables.

Question 4: How similar are objects to each other?

This is particularly important for clustering, recommendation systems and nearest-neighbour algorithms.

Real-world scenario

Suppose a bank wants to identify potentially fraudulent customers.

Before applying clustering, the data scientist needs to:

1. Identify numerical features such as transaction amount.
2. Identify categorical features such as account type.
3. Detect unusually large transactions.
4. Normalize numerical features.
5. Determine how similarity should be calculated between customers.

That is why data understanding comes before data mining.

Source: Uploaded lecture, pages 1–4.

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Page 6: Data objects and attributes

1. What is a data object?

A data object represents an entity about which we collect information.

Other names include:

- Record
- Instance
- Sample
- Example
- Data point

- Tuple

Think of a university student database.

Student ID	Name	Age	Department	CGPA
101	Rahim	23	CSE	3.80
102	Karim	24	EEE	3.65
103	Salma	22	BBA	3.90

Here, each student represents one **data object**.

There are three data objects.

2. What is an attribute?

An attribute is a characteristic or property that describes a data object.

In the previous example:

- Student ID is an attribute.
- Name is an attribute.
- Age is an attribute.
- Department is an attribute.
- CGPA is an attribute.

Other names for attributes include:

Field	Common terminology
Data mining	Attribute
Machine learning	Feature
Statistics	Variable
Data warehousing	Dimension

These terms are often used interchangeably, although their exact meanings can depend on the context.

3. What is an attribute vector?

An attribute vector is the collection of attribute values describing one particular object.

For example:

Rahim:

$$X_1 = (101, 23, CSE, 3.80)$$

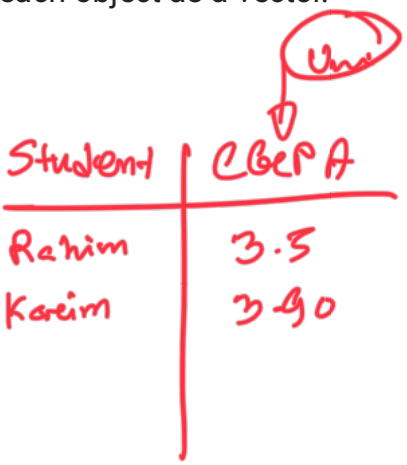
Karim:

$$X_2 = (102, 24, EEE, 3.65)$$

Each vector represents one student using multiple attributes.

This is important because many machine learning algorithms represent each object as a vector.

4. Univariate, bivariate and multivariate data



These terms describe how many attributes you are examining.

Univariate – One attribute

Example: Student CGPA.

$$X = (3.2, 3.5, 3.7, 3.9)$$

You are studying only one variable.

Bivariate – Two attributes

Example: Student study hours and exam scores.

Study hours	Exam score
2	50
4	65
6	80

You can investigate whether study hours are related to exam performance.

Multivariate – Multiple attributes

Example: Predicting student performance using:

- Study hours
- Attendance
- Previous results
- Assignment marks
- Sleep hours

This is a multivariate dataset.

Why is this distinction important?

Different analyses require different numbers of attributes.

- Univariate analysis: Mean, median, histogram.
- Bivariate analysis: Correlation, scatter plot.
- Multivariate analysis: Clustering, dimensionality reduction, multivariate visualization.

Real-world scenario

Suppose you are developing a stock-market manipulation detection system.

For each stock, you collect:

$$X = (Open, High, Low, Close, Volume)$$

This is a multivariate representation because each stock observation contains five attributes.

Your machine learning algorithm will process these attributes to identify unusual trading patterns.

Practice problems

Problem 1: A hospital stores patient ID, age, blood group, weight, height and disease status.

- How many data objects are there if the hospital has 500 patients?
- How many attributes does each object have?
- Is the dataset univariate, bivariate or multivariate?

Answer:

- Data objects = 500 patients.
- Attributes = 6.
- Dataset = Multivariate.

Problem 2: A researcher studies the relationship between temperature and electricity consumption.

What type of analysis is this?

Answer: Bivariate analysis because two attributes are being examined.

Exam tip: Never confuse the number of records with the number of attributes.

Page 7: Nominal attributes

1. What is a nominal attribute?

A nominal attribute represents categories or names that do not have any meaningful order.

Examples:

- Blood group
- Department
- Country
- Occupation
- Operating system
- Stock exchange

Consider:

Student	Department
A	CSE
B	EEE
C	BBA
D	CSE

CSE, EEE and BBA are different categories.

However, there is no logical rule saying:

$$CSE > EEE > BBA$$

These are simply names.

2. What if nominal values are represented by numbers?

This is a very important examination concept.

Suppose:

Department	Encoded value
CSE	1
EEE	2
BBA	3

Does this mean BBA is greater than EEE?

No.

The numbers are only labels.

You cannot calculate:

$$Mean = \frac{1+2+3}{3} = 2$$

And conclude that the average department is EEE.

That would be meaningless.

The numerical encoding does not change the underlying attribute type.

3. What statistical calculation is meaningful?

For nominal attributes, the most appropriate central tendency is the **mode**.

Mode means the most frequently occurring value.

Example:

$$CSE, EEE, CSE, BBA, CSE, EEE$$

Frequency:

Department	Frequency
CSE	3
EEE	2

Department	Frequency
BBA	1

Therefore:

$$Mode = CSE$$

4. Why are nominal attributes used?

They help us classify objects into groups.

For example:

- Divide customers by account type.
- Categorize stocks by industry.
- Classify patients by blood group.
- Group students by academic department.

5. Real-world scenario: Stock-market data

Imagine you have a dataset of 10,000 transactions.

One attribute is `Stock_Sector`.

Possible values:

- Banking
- Pharmaceutical
- Engineering
- Textile
- Energy

You want to know which sector has the highest number of suspicious transactions.

You can count the frequency of each category and identify the mode.

But you cannot calculate the average stock sector.

Practice problems

Problem 1: Consider the following values:

$$Red, Blue, Green, Red, Yellow, Red, Blue$$

Find the mode.

Solution:

- Red = 3
- Blue = 2
- Green = 1
- Yellow = 1

Answer: Red.

Problem 2: A university assigns these codes:

- CSE = 10
- EEE = 20
- BBA = 30

Can you calculate the mean department?

Answer: No. They are nominal labels, not quantitative measurements.

Remember: Nominal means naming or categorizing, not measuring.

Page 7: Binary attributes

1. What is a binary attribute?

A binary attribute is an attribute with only two possible states.

Examples:

- Yes / No
- True / False
- 0 / 1
- Purchased / Not purchased
- Positive / Negative

For example:

Customer	Purchased
A	1

Customer	Purchased
B	0
C	1
D	0

There are only two possible values.

Binary attributes are a special type of nominal attribute.

However, the lecture introduces two important types:

1. Symmetric binary.
2. Asymmetric binary.

2. Symmetric binary attributes

A symmetric binary attribute means both outcomes are equally important.

Example:

Suppose you have a dataset recording whether customers selected one of two equally important options.

Customer	Option A
A	Yes (1)
B	No (0)

Both Yes and No are equally informative for the analysis.

Another commonly used example in introductory data mining is a two-state demographic attribute.

3. Asymmetric binary attributes

An asymmetric binary attribute means one outcome is more important than the other.

Consider disease detection.

Patient	Disease
A	1
B	0
C	0
D	0
E	0

Here:

- 1 = Disease present.
- 0 = Disease absent.

Why is 1 more important?

Because identifying a patient with the disease is much more informative than identifying two patients who both do not have it.

Another example is customer purchasing behaviour.

- 1 = Purchased a product.
- 0 = Did not purchase.

If two customers both did not purchase a product, that does not necessarily mean they have similar preferences.

4. Why do we code the rare outcome as 1?

For asymmetric binary data, the convention is to code the rarer and more important outcome as 1.

For example:

- Fraud = 1.
- Normal transaction = 0.

This makes it easier to focus similarity calculations on meaningful positive events.

5. Real-world scenario: Fraud detection

Imagine a bank has 100,000 transactions.

Only 200 transactions are fraudulent.

If two transactions are both non-fraudulent, that shared negative outcome should not automatically make them highly similar.

Instead, we should focus on shared fraud indicators.

This is why asymmetric binary similarity measures are important.

Practice problems

Problem 1: Identify the binary attribute:

A. Age

B. Blood group

C. Fraud status (Yes/No)

D. Income

Answer: C.

Problem 2: In a disease dataset, which outcome should normally be represented by 1?

Answer: Disease present, because it is the rarer and more important outcome in the asymmetric binary example.

Problem 3: Is purchased/not purchased symmetric or asymmetric when the analysis focuses on purchasing behaviour?

Answer: Asymmetric, because the positive purchase event is more informative.

Page 8: Ordinal attributes

1. What is an ordinal attribute?

An ordinal attribute represents categories that have a meaningful order or ranking.

However, the difference between two consecutive categories is not necessarily equal.

Examples:

- Poor, Fair, Good, Excellent
- Beginner, Intermediate, Advanced
- Low, Medium, High

- Academic ranking
- Customer satisfaction ratings

Consider a customer satisfaction survey:

Customer	Satisfaction
A	Poor
B	Fair
C	Good
D	Excellent

We know:

$$Poor < Fair < Good < Excellent$$

There is a meaningful order.

But can we say that the difference between Poor and Fair is exactly the same as the difference between Good and Excellent?

No. We do not know that.

That is the most important difference between ordinal and numeric attributes.

2. Ordinal vs nominal

Property	Nominal	Ordinal
Categories	Yes	Yes
Meaningful order	No	Yes
Ranking possible	No	Yes
Mean meaningful	No	Generally no
Mode meaningful	Yes	Yes

Property	Nominal	Ordinal
Median meaningful	No	Yes

Example:

- Blood group: Nominal.
- Customer satisfaction: Ordinal.

You can rank satisfaction, but you cannot automatically assume equal distances between satisfaction levels.

3. What statistical measurements can we use?

The lecture identifies:

- Mode
- Median

Example:

$$Poor, Fair, Fair, Good, Excellent$$

The middle value is Fair.

Therefore:

$$Median = Fair$$

Mode is also Fair because it occurs most frequently.

4. Real-world scenario: Product reviews

Suppose an e-commerce platform collects product ratings:

- Very dissatisfied
- Dissatisfied
- Neutral
- Satisfied
- Very satisfied

You want to identify the typical customer satisfaction level.

Using the median is meaningful because the responses have a natural order.

However, treating the categories as equally spaced numerical measurements requires an additional assumption.

5. Practice problems

Problem 1: Classify the following attributes:

1. Employee designation: Junior, Senior, Manager.
2. Blood group: A, B, AB, O.
3. Product quality: Poor, Average, Good, Excellent.
4. Postal code.

Answers:

1. Ordinal.
2. Nominal.
3. Ordinal.
4. Nominal.

Problem 2: Consider:

$$Low, Medium, Medium, High, Very\ High$$

Find the median and mode.

Answer:

- Median = Medium.
- Mode = Medium.

Page 8: Numeric attributes – Interval and ratio

This is one of the most important concepts in the chapter.

Numeric attributes represent measurable quantities.

Unlike nominal and ordinal attributes, numeric attributes support mathematical operations.

However, numeric attributes have two major measurement scales:

1. Interval-scaled.
2. Ratio-scaled.

1. Interval-scaled attributes

An interval attribute has:

- A meaningful order.
- Equal intervals between values.
- No true or absolute zero.

Example: Temperature in Celsius.

Consider:

City	Temperature
Dhaka	20°C
Chattogram	30°C
Sylhet	40°C

We can calculate the differences:

$$30 - 20 = 10$$

$$40 - 30 = 10$$

Both differences represent the same temperature interval.

But can we say 40°C is twice as hot as 20°C?

No.

Why?

Because zero Celsius does not mean the absence of temperature.

The zero point is arbitrary.

Another example is calendar years.

The difference between 2000 and 2010 is 10 years.

But the year 2000 is not twice the year 1000 in any meaningful ratio-scale sense.

2. Ratio-scaled attributes

Ratio attributes have all the properties of interval attributes, plus a meaningful absolute zero.

Examples:

- Age
- Weight
- Height
- Income
- Distance
- Transaction volume
- Duration

Consider:

Person	Monthly income
A	Tk 20,000
B	Tk 40,000
C	Tk 80,000

Now:

$$\frac{80000}{40000} = 2$$

Person C earns twice as much as Person B.

This statement is meaningful because income has a true zero.

3. Interval vs ratio: Why does this matter?

Different mathematical statements become valid depending on the measurement scale.

Operation	Interval	Ratio
Equal / not equal	Yes	Yes
Greater / smaller	Yes	Yes
Addition / subtraction	Yes	Yes
Multiplication / division	Ratios are not meaningful	Yes

Operation	Interval	Ratio
Mean	Yes	Yes
Standard deviation	Yes	Yes

4. Real-world scenario

Suppose you are analysing stock market data.

You collect:

- Stock price.
- Trading volume.
- Trading duration.
- Calendar year.

Classification:

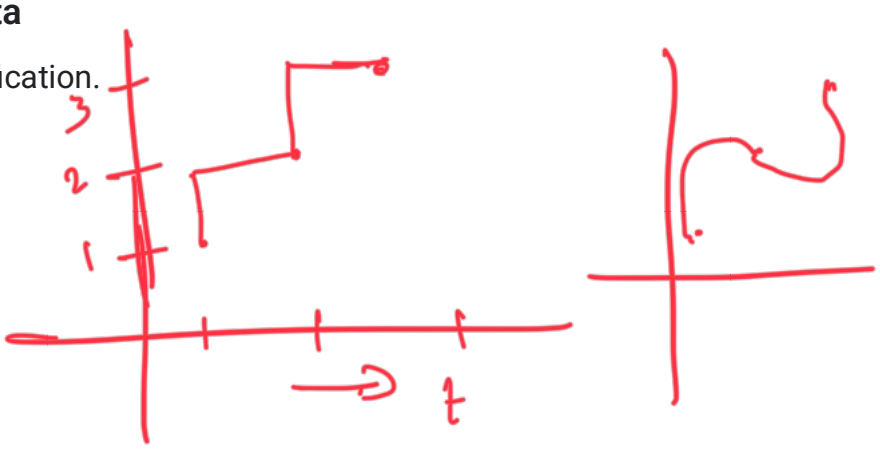
Attribute	Type
Stock price	Ratio
Trading volume	Ratio
Trading duration	Ratio
Calendar year	Interval

You can say that a stock priced at Tk 200 costs twice as much as one priced at Tk 100.

But you cannot say the year 2020 is twice the year 1010.

5. Discrete vs continuous numeric data

The lecture also introduces another classification.



Discrete data

Values are countable.

Examples:

- Number of children.

- Number of transactions.
- Number of employees.
- Number of failed login attempts.

You cannot have 2.5 children or 3.7 employees in a normal counting context.

Continuous data

Values can take any real value within a range.

Examples:

- Height.
- Weight.
- Temperature.
- Time.
- Distance.

For example:

$$Height = 172.56\ cm$$

A continuous variable can have decimal values.

Binary data is a special case of discrete data because it has only two possible values.

Practice problems

Problem 1: Classify the following:

Attribute	Your classification
Temperature in Celsius	?
Monthly salary	?
Number of employees	?
Customer satisfaction level	?
Calendar year	?
Weight	?

Solutions:

Attribute	Classification
Temperature in Celsius	Interval
Monthly salary	Ratio
Number of employees	Discrete numeric
Customer satisfaction	Ordinal
Calendar year	Interval
Weight	Ratio, continuous

Problem 2: Is 50°C twice as hot as 25°C?

Answer: No, because Celsius is interval-scaled.

Problem 3: Is Tk 100,000 twice Tk 50,000?

Answer: Yes, because income is ratio-scaled.

Page 9: Attribute types at a glance

Now let's combine everything.

This page is particularly important for written examinations.

Attribute type	Order	Meaningful operations	Appropriate centre
Nominal	No	Equality / inequality	Mode
Symmetric binary	No	Equality / inequality	Mode
Asymmetric binary	No	Equality / inequality	Mode

Attribute type	Order	Meaningful operations	Appropriate centre
Ordinal	Yes	Equality, ranking	Median, mode
Interval	Yes	Addition, subtraction	Mean, median, mode
Ratio	Yes	Addition, subtraction, multiplication, division	Mean, median, mode

How to identify an attribute type in an examination

Use this decision process:

Step 1: Is it a category or a measurable number?

If category, go to Step 2. If measurable, go to Step 4.

Step 2: Does the category have a natural order?

No → Nominal.

Yes → Ordinal.

Step 3: Does it have only two possible states?

Yes → Binary. Determine whether it is symmetric or asymmetric.

Step 4: Does it have a true zero?

No → Interval.

Yes → Ratio.

Important examination trap

A number does not automatically mean numeric data.

Consider:

- Phone number: Nominal.

- Student ID: Nominal.
- Postal code: Nominal.
- Employee ID: Nominal.

Although these are written using digits, mathematical operations are meaningless.

Practice problems

Classify these 10 attributes:

1. National ID number.
2. Temperature in Kelvin.
3. Customer rating: 1–5 stars.
4. Number of products purchased.
5. Blood group.
6. Monthly income.
7. Product size: Small, Medium, Large.
8. Transaction status: Successful/Failed.
9. Distance travelled.
10. Year of graduation.

Answers:

No.	Answer	Explanation
1	Nominal	Identification label
2	Ratio	Absolute zero
3	Ordinal (in the basic measurement-scale interpretation)	Ordered rating
4	Ratio, discrete	Count with true zero
5	Nominal	Unordered categories
6	Ratio	Meaningful zero

No.	Answer	Explanation
7	Ordinal	Natural order
8	Binary	Two states; symmetry depends on the task
9	Ratio	True zero
10	Interval	Calendar-year differences are meaningful

Part 2: Basic statistical descriptions

Page 11: Central tendency – Mean

1. What is central tendency?

Suppose you have examination scores of 100 students.

You do not want to inspect all 100 scores individually.

Instead, you want to know one value that represents the general performance of the students.

That is the purpose of central tendency.

Central tendency describes the central or typical value of a dataset.

The three most commonly used measurements are:

- Mean.
- Median.
- Mode.

The first topic is mean.

2. Arithmetic mean

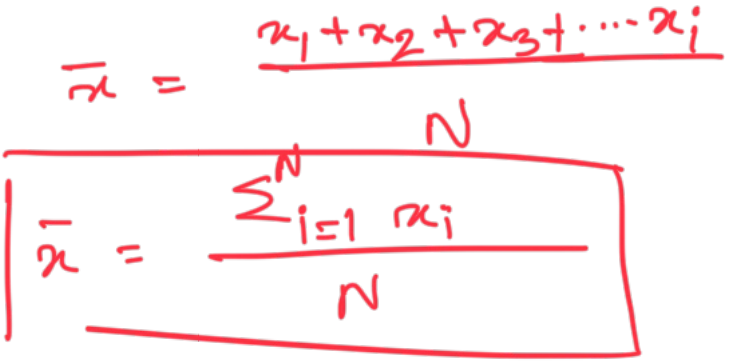
Mean is the sum of all observations divided by the total number of observations.

Formula:

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

Where:

- x_i = Individual observation.
- N = Total number of observations.
- $\sum$ = Summation.
- $\bar{x}$ = Arithmetic mean.



3. Worked example

Suppose five students receive these marks:

$$60, 70, 80, 90, 100$$

Step 1: Add all observations.

$$60 + 70 + 80 + 90 + 100 = 400$$

Step 2: Count the observations.

$$N = 5$$

Step 3: Divide the total by the number of observations.

$$Mean = \frac{400}{5} = 80$$

(Answer: 80)

This means the arithmetic average examination score is 80.

4. Weighted mean

Now imagine your university calculates a course result using two assessments.

Assessment	Marks	Weight
Midterm	80	40%
Final	90	60%

You cannot simply calculate:

$$\frac{80 + 90}{2} = 85$$

Because the assessments have different weights.

Instead:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Substituting:

$$= \frac{(80 \times 0.4) + (90 \times 0.6)}{0.4 + 0.6}$$

$$= 32 + 54$$

$$= 86$$

(Weighted mean = 86)

This is why weighted means are useful when some observations contribute more than others.

5. Why is mean used?

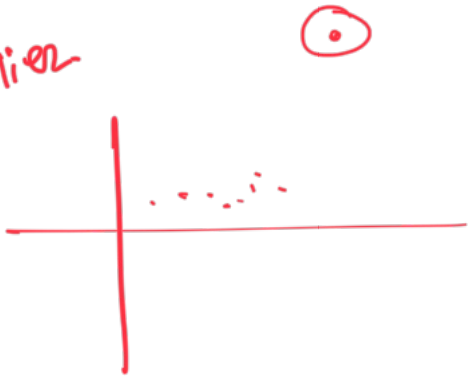
Mean is useful when you want to:

- Calculate average salary.
- Find average stock price.
- Measure average examination performance.
- Calculate average transaction volume.
- Summarize numerical datasets.

6. The biggest weakness of mean: Outliers

Consider:

$$10, 20, 30, 40, 500$$



Mean:

$$\frac{10 + 20 + 30 + 40 + 500}{5} = 120$$

But four out of five values are between 10 and 40.

The mean is 120 because the extreme value 500 pulls the average upward.

This is called sensitivity to outliers.

7. Trimmed mean

A trimmed mean reduces the influence of extreme values.

Suppose:

$$10, 20, 30, 40, 500$$

If we remove the smallest and largest values:

$$20, 30, 40$$

Then:

$$Trimmed\ Mean = \frac{20 + 30 + 40}{3} = 30$$

This better represents the central observations in this example.

The lecture mentions trimming a small percentage from both ends, such as 2%.

8. Distributive, algebraic and holistic measures

This is a slightly more advanced concept from the lecture.

These classifications describe how easily statistical measurements can be computed from partial datasets.

Distributive

Can be calculated by combining partial results.

Examples:

- Sum.
- Count.
- Minimum.
- Maximum.

Suppose three servers process different transactions.

Server A: Sum = 1,000.

Server B: Sum = 2,000.

Server C: Sum = 3,000.

Total:

$$1000 + 2000 + 3000 = 6000$$

No need to collect all original transactions.

Algebraic

Can be calculated from a fixed number of distributive measurements.

Examples:

- Mean.
- Variance.

Mean requires:

$$Mean = \frac{Sum}{Count}$$

Holistic

Generally requires access to the overall distribution or all observations.

Examples:

- Median.
- Mode.

You generally cannot calculate the exact overall median by simply averaging the medians from two servers.

Real-world scenario

A stock exchange processes millions of transactions every day.

To calculate average daily transaction value, each server can send its total transaction value and transaction count.

The central system combines them.

This is an algebraic aggregation approach.

Practice problems

Problem 1: Find the mean:

$$12, 18, 20, 25, 30$$

Solution:

$$Mean = \frac{12 + 18 + 20 + 25 + 30}{5}$$

$$= \frac{105}{5} = 21$$

Problem 2: Find the weighted mean:

Subject	Marks	Weight
Mathematics	80	30%
Programming	90	50%
Database	70	20%

Solution:

$$= (80 \times 0.3) + (90 \times 0.5) + (70 \times 0.2)$$

$$= 24 + 45 + 14 = 83$$

Problem 3: Which measure is more sensitive to outliers, mean or median?

Answer: Mean.

Page 12: Median, mode and midrange

1. Median

Median is the middle value of a sorted dataset.

The first step is always to sort the values.

Case 1: Odd number of observations

Example:

$$10, 20, 30, 40, 50$$

There are five observations.

The middle value is:

$$Median = 30$$

Position:

$$\frac{N+1}{2}$$

$$= \frac{5+1}{2} = 3$$

The third observation is 30.

Case 2: Even number of observations

Example:

$$10, 20, 30, 40, 50, 60$$

There are six observations.

There is no single middle observation.

So, take the average of the two middle values.

$$Median = \frac{30+40}{2} = 35$$

Position:

$$\frac{N}{2} = 3$$

And:

$$\frac{N}{2} + 1 = 4$$

Average the third and fourth observations.

2. Why is median useful?

Median is robust against outliers.

Consider monthly salaries:

$$15000, 18000, 20000, 22000, 500000$$

Mean:

$$= \frac{575000}{5} = 115000$$

Median:

$$= 20000$$

The mean suggests that the typical salary is Tk 115,000, even though most people earn around Tk 15,000–22,000.

Median gives a more representative central value here.

Real-world applications:

- House prices.
- Salaries.
- Hospital waiting times.
- Income distribution.
- Transaction values with extreme observations.

3. Mode

Mode is the most frequently occurring observation.

Example:

$$10, 20, 20, 30, 30, 30, 40, 50$$

Frequency:

Value	Frequency
10	1
20	2
30	3
40	1
50	1

Therefore:

$$Mode = 30$$

A dataset can have different modal structures:

- Unimodal: One mode.
- Bimodal: Two modes.
- Trimodal: Three modes.
- Multimodal: Multiple modes.

Example:

$$10, 10, 20, 20, 30$$

Modes:

$$10, 20$$

This is bimodal.

Why is mode useful?

Suppose an online shop wants to identify its most popular product size.

The mode identifies the most frequently selected size.

It also works with nominal data, where mean and median are not appropriate.

4. Midrange

Midrange is the average of the minimum and maximum observations.

Formula:

$$Midrange = \frac{Min + Max}{2}$$

Example:

$$10, 20, 30, 40, 50$$

$$Midrange = \frac{10 + 50}{2} = 30$$

It is easy to calculate, but very sensitive to extreme values.

5. Comparing all four measurements

Consider:

$$10, 20, 30, 30, 40, 100$$

Measure	Calculation	Result
Mean	230 / 6	38.33
Median	$30 + 30) / 2	30
Mode	Most frequent	30

Measure	Calculation	Result
Midrange	(10 + 100) / 2	55

Notice that each measure tells us something slightly different.

6. Grouped data – Finding median

Sometimes you don't have individual observations. Instead, you have a frequency table.

Example:

Age group	Frequency	Cumulative frequency
0–20	120	120
20–40	300	420
40–60	260	680
60–80	120	800

Total:

$$N = 800$$

Step 1: Find the median position.

$$N/2 = 400$$

Step 2: Find the class containing the 400th observation.

Cumulative frequency reaches 420 in the 20–40 group.

Therefore, median class = 20–40.

Step 3: Apply the grouped median formula.

$$Median = L_1 + \frac{N/2 - CF}{f} \times h$$

Where:

- (L_1=20$: Lower boundary of median class.
- $CF = 120$: Cumulative frequency before median class.

- $f = 300$: Frequency of median class.
- $h = 20$: Class width.
- $N = 800$.

Substitute:

$$Median = 20 + \frac{400 - 120}{300} \times 20$$

$$= 20 + \frac{280}{300} \times 20$$

$$= 20 + 18.67$$

$$\boxed{Median \approx 38.67}$$



This is an estimate based on interpolation within the median class.

Practice problems

Problem 1: Find the median:

$$5, 10, 15, 20, 25, 30, 35$$

Answer:

$$Median = 20$$

Problem 2: Find the median:

$$12, 18, 22, 25, 30, 35$$

Answer:

$$Median = \frac{22 + 25}{2} = 23.5$$

Problem 3: Find the mode:

$$2, 3, 3, 4, 5, 5, 5, 6, 7$$

Answer: 5.

Problem 4: Find the midrange:

$$15, 20, 25, 30, 45$$

Answer:

$$\frac{15 + 45}{2} = 30$$

Page 13: Symmetry and skewness

1. What is skewness?

Skewness describes whether a distribution is symmetrical or stretched more towards one side.

Imagine you have a histogram of 1,000 people's monthly income.

Most people earn between Tk 20,000 and Tk 50,000, but a few earn Tk 500,000 or more.

The distribution will have a long tail towards higher values.

That is positive skewness.

2. Symmetrical distribution

In a perfectly symmetrical, unimodal distribution:

$$Mean = Median = Mode$$

Example:

$$10, 20, 30, 40, 50$$

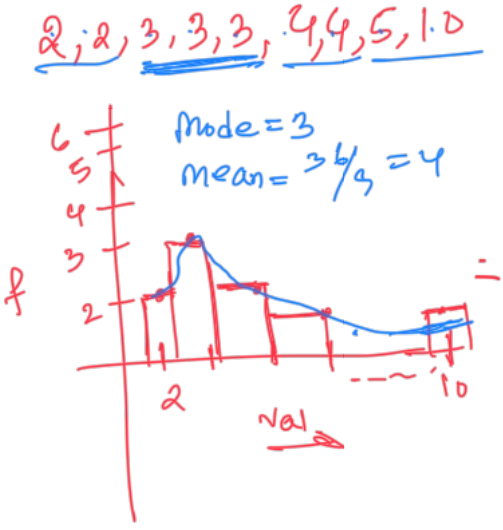
Mean = 30.

Median = 30.

Mode is not unique in this particular dataset, so it is not a perfect example of a unimodal distribution.
The important idea is that a symmetric unimodal distribution has its centre aligned.

3. Positively skewed distribution (right-skewed)

In a positively skewed distribution, the tail extends towards the right.



Typical relationship:

$$Mode < Median < Mean$$

Example: Income distribution.

Suppose:

$$10, 15, 20, 25, 30, 100$$

Mean:

$$\frac{200}{6} = 33.33$$

Median:

$$\frac{20 + 25}{2} = 22.5$$

The high value 100 pulls the mean towards the right.

4. Negatively skewed distribution (left-skewed)

In a negatively skewed distribution, the tail extends towards the left.

Typical relationship:

$$Mean < Median < Mode$$

Example: An examination where most students achieve high marks, but a few perform very poorly.

The low scores pull the mean towards the left.

5. Why is skewness important?

Because the mean alone can give a misleading picture of a dataset.

For example, when analysing household income in Bangladesh:

- Mean income may be affected by very wealthy households.
- Median income may better represent the middle household.
- Skewness reveals the direction of the long tail.

Skewness also influences decisions about preprocessing, statistical analysis and model selection.

6. Pearson's empirical relationship

The lecture gives this approximate relationship for moderately skewed distributions:

$$Mean - Mode \approx 3(Mean - Median)$$

This is an empirical approximation, not an exact mathematical identity.

Practice problems

Problem 1: If:

- Mean = 80
- Median = 65
- Mode = 55

Identify the skewness.

Answer:

$$Mode < Median < Mean$$

Therefore, positively skewed.

Problem 2: If:

- Mean = 40
- Median = 50
- Mode = 60

Answer: Negatively skewed.

Problem 3: A salary dataset contains mostly low and medium salaries with a few extremely high salaries. What type of skewness is expected?

Answer: Positive skewness.

Page 14: Worked example – Central tendency

This page combines the previous concepts.

Given examination scores of 12 students:

$$42, 48, 55, 58, 61, 61, 64, 67, 70, 72, 75, 98$$

We need to calculate:

1. Mean.
2. Median.
3. Mode.
4. Midrange.
5. Trimmed mean.

Step 1: Mean

$$Mean = \frac{\sum x}{N}$$

Total:

$$42 + 48 + 55 + 58 + 61 + 61 + 64 + 67 + 70 + 72 + 75 + 98 = 771$$

$$Mean = \frac{771}{12}$$

$$\boxed{Mean = 64.25}$$

Step 2: Median

There are 12 observations, which is even.

Therefore:

$$Median = \frac{x_6 + x_7}{2}$$

$$= \frac{61 + 64}{2}$$

$$\boxed{Median = 62.5}$$

Step 3: Mode

The value 61 appears twice.

All other values appear once.

$$\boxed{Mode = 61}$$

Step 4: Midrange

Minimum = 42.

Maximum = 98.

$$Midrange = \frac{42 + 98}{2}$$

$$\boxed{Midrange = 70}$$

Step 5: Trimmed mean

Remove one smallest and one largest value.

Original:

$$42, 48, 55, 58, 61, 61, 64, 67, 70, 72, 75, 98$$

After removing 42 and 98:

$$48, 55, 58, 61, 61, 64, 67, 70, 72, 75$$

New total:

$$771 - 42 - 98 = 631$$

New number of observations:

$$10$$

$$Trimmed\ Mean = \frac{631}{10}$$

$$\boxed{63.1}$$

Step 6: Interpret the results

Measure	Result
Mean	64.25
Median	62.5
Mode	61
Midrange	70
Trimmed mean	63.1

Notice:

$$Mode < Median < Mean$$

This suggests mild positive skewness.

Why?

Because the unusually high score of 98 pulls the mean upward.

The trimmed mean falls from 64.25 to 63.1 after removing the two extremes.

Why does this example matter?

In a real dataset, you should not just calculate the average and stop.

You should interpret the measurements.

The differences between mean, median and mode tell you something about the distribution.

Practice problem

Consider:

$$20, 25, 30, 35, 40, 45, 50, 100$$

Calculate:

1. Mean.
2. Median.
3. Mode.
4. Midrange.
5. Trimmed mean after removing one value from each end.
6. Identify the likely skewness.

(Solutions)

- Mean = 43.125.
- Median = 37.5.
- Mode = No unique mode.
- Midrange = 60.
- Trimmed mean = 35.83.
- Likely positively skewed.

Page 16: Dispersion – Quantiles, IQR and five-number summary

1. What is dispersion?

Central tendency tells you where the centre of the data is.

Dispersion tells you how spread out the data is.

Consider two classrooms:

Class A:

$$48, 49, 50, 51, 52$$

Class B:

$$10, 30, 50, 70, 90$$

Both have mean:

$$50$$

But are they similar?

No.

Class A's marks are tightly grouped.

Class B's marks are widely distributed.

This is why we need dispersion measurements.

2. Range

Range is the difference between the maximum and minimum values.

$$Range = Max - Min$$

Class A:

$$52 - 48 = 4$$

Class B:

$$90 - 10 = 80$$

Class B has much greater spread.

Why use range?

It gives a quick understanding of the total spread.

Limitation: It depends only on two observations and is sensitive to outliers.

3. Quantiles

Quantiles divide sorted observations into equal-sized groups.

The lecture discusses:

- 2-quantiles.
- 4-quantiles.
- 100-quantiles.

2-quantiles: Divide data into two equal parts.



This is the median.

4-quantiles: Divide data into four equal parts.



These are the quartiles:

- Q1 = First quartile (25th percentile).
- Q2 = Second quartile (50th percentile / median).
- Q3 = Third quartile (75th percentile).

100-quantiles: Percentiles.



For example, the 90th percentile indicates a value at or below which approximately 90% of observations fall, subject to the chosen quantile convention.

4. Interquartile range (IQR)

IQR measures the spread of the middle 50% of observations.

Formula:

$$IQR = Q3 - Q1$$

Example:

$$Q1 = 20, \quad Q3 = 60$$

$$IQR = 60 - 20 = 40$$

This tells us that the middle half of the observations spans 40 units.

Why is IQR useful?

Unlike range, IQR is not directly affected by the smallest and largest observations.

It is useful when a dataset contains outliers.

5. Five-number summary

The five-number summary gives a compact description of a dataset.

It consists of:

1. Minimum.
2. Q1.
3. Median.
4. Q3.
5. Maximum.

Example:

$$10, 20, 30, 40, 50, 60, 70, 80, 90$$

Using the median-of-halves convention:

- Minimum = 10.
- Q1 = 25.
- Median = 50.
- Q3 = 75.
- Maximum = 90.

Therefore:

$$(10, 25, 50, 75, 90)$$

6. Outlier detection using IQR

This is extremely important.

The lecture uses the 1.5 × IQR rule.

Lower fence:

$$LF = Q1 - 1.5(IQR)$$

Upper fence:

$$UF = Q3 + 1.5(IQR)$$

Any observation outside these fences is a suspected outlier.

Example:

$$Q1 = 25, \quad Q3 = 75$$

$$IQR = 75 - 25 = 50$$

Lower fence:

$$25 - 1.5(50) = -50$$

Upper fence:

$$75 + 1.5(50) = 150$$

Any value below -50 or above 150 is a suspected outlier.

Important: An outlier is not automatically an error. It may be a genuine rare observation.

7. Real-world scenario: Detecting abnormal stock transactions

Suppose you analyse daily transaction volumes:

$$100, 120, 130, 140, 150, 160, 170, 180, 200, 1000$$

The transaction volume of 1000 may represent unusual trading activity.

You can use the IQR method to investigate whether it is an outlier.

But do not automatically delete it. It may be exactly the suspicious market activity you want to detect.

8. Quartile convention warning

The lecture uses the **median-of-halves** method.

Different software packages may calculate quartiles differently.

For example, Python, Excel and statistical packages can use different quantile interpolation methods.

In examinations, follow the convention specified in the question or lecture.

Practice problems

Problem 1: Find the range:

$$12, 15, 18, 22, 30, 45$$

Answer:

$$45 - 12 = 33$$

Problem 2: Given:

$$Q1 = 30, \quad Q3 = 70$$

Calculate IQR and outlier fences.

Answer:

$$IQR = 40$$

$$LF = 30 - 60 = -30$$

$$UF = 70 + 60 = 130$$

Problem 3: Is 150 a suspected outlier?

Yes, because it exceeds the upper fence of 130.

Page 17: Boxplot

1. What is a boxplot?

A boxplot is a graphical representation of the five-number summary.

It helps us understand:

- Minimum.
- First quartile.
- Median.
- Third quartile.
- Maximum.
- Suspected outliers.

2. Understanding each component

The box

The box extends from Q1 to Q3.

It represents the middle 50% of observations.

The line inside the box

This represents the median.

The whiskers

These extend to the smallest and largest non-outlier observations.

Individual points beyond the whiskers

These represent suspected outliers.

3. Worked example from the lecture

Examination scores:

$$42, 48, 55, 58, 61, 61, 64, 67, 70, 72, 75, 98$$

(Solved in the lecture)

We already know:

$$Median = 62.5$$

Now calculate Q1.

Lower half:

$$42, 48, 55, 58, 61, 61$$

$$Q1 = \frac{55 + 58}{2} = 56.5$$

Calculate Q3.

Upper half:

$$64, 67, 70, 72, 75, 98$$

$$Q3 = \frac{70 + 72}{2} = 71$$

Calculate IQR:

$$IQR = 71 - 56.5 = 14.5$$

Lower fence:

$$56.5 - 1.5(14.5)$$

$$= 34.75$$

Upper fence:

$$71 + 1.5(14.5)$$

$$= 92.75$$

Now check each observation.

The value 98 is greater than 92.75.

Therefore:

$$\boxed{98 \text{ is a suspected outlier}}$$

The largest non-outlier is 75, so the upper whisker ends at 75.

Five-number summary:

$$42, 56.5, 62.5, 71, 98$$

4. Why are boxplots used?

Imagine you want to compare salary distributions across four departments.

Instead of creating four large tables, you can create four boxplots.

You can quickly identify:

- Which department has the highest median salary.

- Which department has the greatest variation.
- Which department has unusually high salaries.
- Which department has a more concentrated salary distribution.

5. Real-world scenario

In your stock-market research, you could create boxplots of daily percentage price changes for different stocks.

This would help identify stocks with unusual volatility or extreme price movements.

Practice problems

Given:



$$5, 10, 15, 20, 25, 30, 35, 40, 45, 100$$

Using the median-of-halves convention, calculate:

1. Q1.
2. Median.
3. Q3.
4. IQR.
5. Lower fence.
6. Upper fence.
7. Suspected outlier.

Solution:

Lower half:

$$5, 10, 15, 20, 25$$

$$Q1 = 15$$

Median:

$$Median = \frac{25 + 30}{2} = 27.5$$

Upper half:

$$30, 35, 40, 45, 100$$

$$Q3 = 40$$

IQR:

$$40 - 15 = 25$$

Lower fence:

$$15 - 1.5(25) = -22.5$$

Upper fence:

$$40 + 1.5(25) = 77.5$$

Since:

$$100 > 77.5$$

100 is a suspected outlier.

Page 18: Variance and standard deviation

This is another very important mathematical topic.

1. Why do we need variance and standard deviation?

Suppose two students receive these examination scores:

Student A:

$$48, 49, 50, 51, 52$$

Student B:

$$10, 30, 50, 70, 90$$

Both have the same mean:

$$Mean = 50$$

But Student B's results are much more spread out.

Mean cannot explain this difference.

Variance and standard deviation measure how far observations are spread around the mean.

2. Variance

Variance calculates the average squared deviation from the mean.

Population variance formula:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N}$$

Where:

- x_i = Individual value.
- $\bar{x}$ = Mean.
- N = Number of observations.
- σ^2 = Population variance.

Why square the differences?

Because some observations are above the mean and others are below it.

Without squaring, positive and negative deviations could cancel each other.

3. Step-by-step worked example

Consider:

$$2, 4, 6, 8, 10$$

Step 1: Calculate the mean.

$$Mean = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Step 2: Calculate deviations.

Value	Value – Mean
2	-4
4	-2
6	0
8	2
10	4

Step 3: Square each deviation.

Value	Deviation	Squared deviation
2	-4	16
4	-2	4
6	0	0
8	2	4
10	4	16

Step 4: Sum squared deviations.

$$16 + 4 + 0 + 4 + 16 = 40$$

Step 5: Divide by N.

$$\sigma^2 = \frac{40}{5}$$

$$\boxed{\sigma^2 = 8}$$

4. Standard deviation

Standard deviation is the square root of variance.

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{8}$$

$$\boxed{\sigma \approx 2.83}$$

Why do we need standard deviation?

Variance is measured in squared units.

If the original data represents marks, variance represents squared marks.

Standard deviation returns the measurement to the original unit.

Therefore, standard deviation is usually easier to interpret.

5. Interpreting standard deviation

- Small standard deviation: Observations are close to the mean.

- Large standard deviation: Observations are spread out.
- Zero standard deviation: All observations are identical.

6. Population vs sample variance

This distinction frequently appears in examinations.

Population variance:

$$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{N}$$

Sample variance:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{N - 1}$$

Why divide by N−1 for a sample?

Because the sample mean is estimated from the sample itself. Dividing by N−1 corrects the tendency of the sample variance to underestimate population variance.

For the previous example:

Population variance:

$$\sigma^2 = \frac{40}{5} = 8$$

Sample variance:

$$s^2 = \frac{40}{4} = 10$$

Sample standard deviation:

$$s = \sqrt{10} \approx 3.16$$

Exam warning: Your lecture uses N for its displayed variance calculations, but questions may explicitly ask for sample variance. Always check which one is required.

7. Shortcut formula

The lecture also provides:

$$\sigma^2 = \frac{\sum x_i^2}{N} - \bar{x}^2$$

This can simplify calculations.

For:

$$2, 4, 6, 8, 10$$

Squares:

$$4, 16, 36, 64, 100$$

Sum:

$$220$$

$$\sigma^2 = \frac{220}{5} - 6^2$$

$$= 44 - 36 = 8$$

Same answer.

8. Chebyshev's inequality

The lecture introduces an important statistical rule.

At least:

$$1 - \frac{1}{k^2}$$

of observations lie within k standard deviations of the mean for $k > 1$.

This applies regardless of the distribution's shape, provided the variance is finite.

For $k = 2$:

$$1 - \frac{1}{2^2}$$

$$= 1 - \frac{1}{4} = 0.75$$

Therefore, at least 75% of observations lie within two standard deviations of the mean.

For $k = 3$:

$$1 - \frac{1}{9} = \frac{8}{9}$$

Approximately 88.89% lie within three standard deviations.

9. Real-world scenario: Stock volatility

Suppose two stocks have these daily returns:

Stock A:

$$1\%, 2\%, 1.5\%, 2.2\%, 1.8\%$$

Stock B:

$$-8\%, 5\%, 12\%, -6\%, 10\%$$

Stock B has much greater variability.

Standard deviation helps quantify this volatility.

It is useful in:

- Financial risk analysis.
- Quality control.
- Anomaly detection.
- Machine learning preprocessing.
- Comparing consistency between datasets.

Practice problems

Problem 1: Calculate population variance and standard deviation:

$$4, 6, 8, 10, 12$$

Solution:

Mean:

$$\bar{x} = 8$$

Squared deviations:

$$16, 4, 0, 4, 16$$

Variance:

$$\sigma^2 = \frac{40}{5} = 8$$

Standard deviation:

$$\sigma = \sqrt{8} = 2.83$$

Problem 2: A dataset has mean 100 and standard deviation 10.

According to Chebyshev's inequality, at least what percentage lies between 80 and 120?

Answer:

Two standard deviations:

$$k = 2$$

At least:

$$1 - \frac{1}{4} = 75\%$$

Problem 3: If variance is 0, what does it mean?

Answer: All observations have exactly the same value.

Part 3: Data visualization

Page 19: Graphic displays of basic statistics

So far, we have learned how to describe data mathematically.

But imagine you have 50,000 records.

Would you manually inspect every value to understand the distribution?

Obviously not.

This is where data visualization becomes important.

Data visualization transforms numerical information into graphical representations so humans can identify patterns more easily.

The lecture introduces four important graphical displays:

1. Histogram.
2. Quantile plot.
3. Q-Q plot.
4. Scatter plot.

1. Histogram

A histogram displays the frequency distribution of numerical data using adjacent bars.

It divides numerical values into intervals called bins or buckets.

Example: Examination scores.

Score range	Number of students
0–20	2
20–40	5
40–60	12
60–80	20
80–100	11

The horizontal axis represents score ranges.

The vertical axis represents frequency.

For example, the 60–80 interval contains 20 students.

Why is a histogram used?

It helps identify:

- Distribution shape.
- Concentration of observations.
- Skewness.
- Multiple peaks.
- Gaps.
- Potentially unusual observations.

Important difference: Histogram vs bar chart

Histogram	Bar chart
Numerical continuous or grouped data	Categorical data
Bars normally touch	Bars normally have gaps
Order follows numerical intervals	Order follows categories or chosen arrangement
Shows distribution	Compares category frequencies

Example:

- Histogram: Distribution of customer ages.
- Bar chart: Number of customers by department.

Why does bin width matter?

Suppose you have 1,000 transaction values.

If you use only two large bins, important patterns may disappear.

If you use 100 very small bins, the graph may become noisy.

Therefore, selecting a reasonable bin width is important.

2. Quantile plot

A quantile plot displays sorted observations against their corresponding cumulative fractions.

The lecture uses:

$$f_i = \frac{i - 0.5}{N}$$

Where:

- i = Position of the sorted observation.
- N = Total number of observations.

Example:

$$10, 20, 30, 40, 50$$

Here:

$$N = 5$$

For the first value:

$$f_1 = \frac{1 - 0.5}{5} = 0.1$$

For the second value:

$$f_2 = \frac{2 - 0.5}{5} = 0.3$$

The complete table:

Position	Value	Quantile fraction
1	10	0.1
2	20	0.3
3	30	0.5
4	40	0.7
5	50	0.9

Why is it used?

A quantile plot allows you to see the distribution of all observations, rather than only the mean or median.

It helps identify:

- Where observations are concentrated.
- Whether values increase gradually or suddenly.
- Whether the distribution has long tails.
- Unusually large observations.

3. Q-Q plot (Quantile-Quantile plot)

This is a very important visualization concept.

A Q-Q plot compares the quantiles of two distributions.

For example:

You have:

- Distribution A: Actual examination scores.
- Distribution B: Normally distributed theoretical values.

You want to know whether the examination scores approximately follow a normal distribution.

A Q-Q plot can help you investigate this.

Interpretation:

- Points approximately following a 45-degree line: The distributions agree reasonably well.

- Curved pattern: The distributions differ systematically.
- Large deviations at the ends: Their tails differ.
- Individual distant points: Possible unusual observations.

Why is it used?

Many statistical techniques make assumptions about the distribution of data.

Q-Q plots help investigate those assumptions.

4. Scatter plot

A scatter plot displays the relationship between two numerical attributes.

Each point represents one data object.

Example:

Study hours	Exam score
1	40
2	45
3	55
4	65
5	75
6	85

Each point has two coordinates:

$$(x, y)$$

For example:

$$(3, 55)$$

Means the student studied for three hours and scored 55.

Why is it used?

It helps discover:

- Positive relationships.
- Negative relationships.
- Clusters.
- Outliers.
- Possible nonlinear relationships.

Real-world scenario

Suppose you want to analyse whether trading volume is related to daily stock price changes.

X-axis:

$$Trading\ Volume$$

Y-axis:

$$Daily\ Price\ Change$$

A scatter plot can reveal whether unusually high trading volume is associated with large price movements.

However, correlation does not automatically establish causation.

Practice problems

Problem 1: Which visualization should you use?

Situation	Answer
Distribution of employee salaries	?
Relationship between age and salary	?
Compare actual data against theoretical normal distribution	?
Display every sorted observation against cumulative fraction	?
Number of employees in each department	?

Answers:

1. Histogram.
2. Scatter plot.

3. Q-Q plot.
4. Quantile plot.
5. Bar chart.

Problem 2: A scatter plot shows points rising from left to right. What does this suggest?

Answer: Positive correlation.

Page 20: Reading correlation from scatter plots

1. What is correlation?

Correlation describes how two variables change in relation to each other.

Suppose you are studying:

- Study hours.
- Examination scores.

When study hours increase, examination scores may also increase.

This indicates positive correlation.

2. Positive correlation

When one variable increases, the other generally increases.



Examples:

- Study hours and exam performance.
- Advertising expenditure and sales, in some settings.
- Height and weight in many populations.

3. Negative correlation

When one variable increases, the other generally decreases.

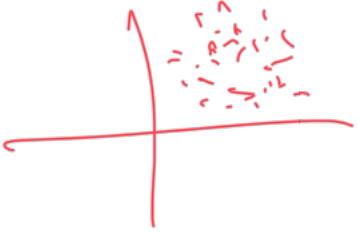


Examples:

- Product price and demand, under suitable market conditions.
- Distance remaining and journey completion percentage.
- Interest rate and borrowing demand, in some settings.

4. No linear correlation

There is no clear straight-line relationship.



This does not necessarily mean there is no relationship at all.

For example:

$$y = x^2$$

may show a strong nonlinear relationship even though a linear correlation measure could be close to zero.

5. Pearson correlation coefficient

The lecture mentions Pearson's correlation coefficient, although formal computation is introduced later.

Its value ranges from:

$$-1 \leq r \leq 1$$

Interpretation:

Value	Interpretation
+1	Perfect positive linear correlation
0	No linear correlation
-1	Perfect negative linear correlation

Real-world scenario

You work with a stock brokerage company.

You want to understand whether daily trading volume and daily percentage price change move together.

A scatter plot is your first exploratory tool.

If you observe a pattern, you can investigate it using correlation analysis.

Practice problems

Problem 1: Identify the relationship:

- X increases, Y increases.

- X increases, Y decreases.
- X increases, Y has no obvious linear pattern.

Answers:

- Positive.
- Negative.
- No linear correlation.

Problem 2: What does $r = -0.95$ indicate?

Answer: A very strong negative linear relationship.

Problem 3: Does $r = 0$ prove that two variables are completely unrelated?

Answer: No. They may have a nonlinear relationship.

Page 22: Four families of multidimensional visualization

1. Why do we need multidimensional visualization?

So far, a scatter plot has allowed us to visualize two attributes.

But what happens when a dataset contains 20 attributes?

For example, a customer dataset might contain:

1. Age.
2. Income.
3. Education.
4. Location.
5. Spending.
6. Purchase frequency.
7. Credit score.
8. Account duration.
9. Number of transactions.
10. Product preferences.
11. And more.

Visualizing all these attributes simultaneously is difficult.

Multidimensional visualization techniques help represent many dimensions in a visual format.

The lecture identifies four families.

2. Pixel-oriented visualization

Core idea: Represent individual data values using coloured pixels.

Each attribute is assigned a visualization window, and values are represented using colours.

Imagine a dataset with one million stock transactions.

Instead of displaying one million rows in a table, you encode transaction values using different colours.

A viewer can identify dense patterns, unusual regions and broad structures.

Why is it used?

- Handles very large datasets.
- Uses screen space efficiently.
- Helps reveal large-scale patterns.

Limitation: Exact numerical values are difficult to read from colours, and colour selection matters.

Scenario: Visualizing millions of network traffic records to find unusual activity.

3. Geometric projection techniques

Core idea: Project multidimensional data into two-dimensional or three-dimensional visual layouts.

Examples:

- Scatter-plot matrix.
- Parallel coordinates.

A scatter-plot matrix shows relationships between pairs of attributes.

Suppose you have four attributes:

- Age.
- Income.
- Spending.
- Credit score.

A scatter-plot matrix creates pairwise visualizations.

Why is it used?

To investigate relationships between multiple attributes.

Limitation: The number of plots increases quickly as dimensions increase.

For n attributes, there are:

$$\frac{n(n-1)}{2}$$

unique pairwise comparisons.

For 10 attributes:

$$\frac{10(9)}{2} = 45$$

That means 45 pairwise comparisons!

4. Icon-based visualization

Core idea: Represent each data object using an icon.

Each attribute controls a visual characteristic of the icon.

One example from the lecture is Chernoff faces.

For example:

- Face width represents income.
- Eye size represents age.
- Mouth curvature represents satisfaction.
- Nose length represents another attribute.

You can visually compare objects by comparing their faces.

Why is it used?

Humans are good at identifying differences in familiar visual patterns.

Limitation: Different facial features attract different levels of attention, so interpretation can be subjective.

Scenario: Comparing the profiles of different customer groups.

5. Hierarchical visualization

Core idea: Organize data into nested groups or levels.

Examples:

- Treemaps.
- Worlds-within-worlds.
- Dimensional stacking.

Imagine a company structure:

Company → Department → Team → Employee.

A treemap can represent these nested relationships using rectangles.

The size or colour of each rectangle can represent another measurement.

Why is it used?

- Shows hierarchical structures.
- Helps visualize category proportions.
- Makes nested relationships easier to understand.

Scenario: Visualizing a company's revenue by business unit, department and product.

6. Complex data and relationships

The lecture also mentions:

Tag clouds

Used for text data.

Frequently occurring words appear larger.

Node-link graphs

Used for relationships between entities.

Examples:

- Social networks.
- Citation networks.
- Communication networks.

Comparison table

Visualization	Main idea	Best use
Pixel-oriented	Colour pixels	Massive datasets
Geometric projection	2D/3D projections	Attribute relationships
Icon-based	Encode attributes into icons	Comparing objects visually
Hierarchical	Nested structures	Parent-child relationships

Practice problems

Problem 1: Which visualization is appropriate?

1. Visualize millions of network records using coloured pixels.
2. Compare 10 numerical features using pairwise plots.
3. Represent customers using faces.
4. Display company revenue by nested departments.
5. Display social network connections.

Answers:

1. Pixel-oriented.
2. Geometric projection.
3. Icon-based.
4. Hierarchical.
5. Node-link graph.

Problem 2: A company wants to display sales for products, categories and subcategories using nested rectangles. Which technique?

Answer: Treemap, a hierarchical visualization.

Page 23: Parallel coordinates

1. What are parallel coordinates?

Parallel coordinates are a multidimensional visualization technique.

Instead of representing attributes as the X and Y coordinates of a normal scatter plot, we create multiple parallel vertical axes.

Each axis represents one attribute.

Each data object becomes a line connecting its values across all axes.

2. Understand using an example

Suppose you have three students.

Student	Mathematics	Programming	Database
A	90	85	95
B	40	45	50
C	70	75	80

You create three vertical axes:

- Mathematics.
- Programming.
- Database.

Each student is represented by a line.

Student A's line connects:

$$90 \rightarrow 85 \rightarrow 95$$

Student B's line connects:

$$40 \rightarrow 45 \rightarrow 50$$

Student C's line connects:

$$70 \rightarrow 75 \rightarrow 80$$

The lines help you visually identify groups with similar performance patterns.

3. Why is it used?

A normal scatter plot can represent only two numerical attributes at once.

Parallel coordinates allow you to inspect many attributes simultaneously.

Applications:

- Financial risk analysis.
- Customer segmentation.
- Healthcare data.
- Network monitoring.
- Manufacturing quality control.

4. What is the limitation?

Too many observations can produce clutter.

Imagine 100,000 transactions.

All their lines may overlap and become difficult to interpret.

This is called overplotting or visual clutter.

5. Real-world scenario

Suppose you are analysing suspicious trading activities using:

- Price change.
- Volume.
- Volatility.
- Transaction frequency.
- Order cancellation ratio.

Parallel coordinates can help you visually compare normal and suspicious transaction patterns across all five attributes.

Practice problem

A dataset contains five features:

$$Age, Income, Spending, Visits, Purchase$$

How many parallel axes will you need?

Answer: Five axes.

What represents one customer?

Answer: One polyline connecting the customer's values across the five axes.

Part 4: Similarity and dissimilarity

Page 25: Proximity and the two data structures

This is one of the most important sections in the entire chapter because it connects data understanding to clustering and machine learning.

1. What is similarity?

Similarity measures how alike two objects are.

Higher similarity means objects are more alike.

For example:

- Two customers with similar shopping behaviour.
- Two stocks with similar trading patterns.
- Two documents discussing similar subjects.

A similarity score often ranges between 0 and 1.

$$0 \leq Similarity \leq 1$$

Where:

- 1 = Identical or maximally similar.
- 0 = Completely dissimilar under that measure.

2. What is dissimilarity?

Dissimilarity measures how different two objects are.

A smaller dissimilarity indicates greater resemblance.

For example:

- Distance between two customers.
- Difference between two stock price patterns.
- Difference between two documents.

A common normalized relationship is:

$$sim(i, j) = 1 - d(i, j)$$

This relationship applies when dissimilarity has been normalized to the range $0, 1$.

Example:

If:

$$d(A, B) = 0.25$$

Then:

$$sim(A, B) = 1 - 0.25 = 0.75$$

So the similarity is 0.75.

3. Data matrix (two-mode)

A data matrix stores objects against attributes.

Example:

Customer	Age	Income	Purchases
A	25	30000	5
B	30	50000	8
C	40	70000	10

Here:

- Rows = Objects.
- Columns = Attributes.

If there are n objects and p attributes, the matrix has:

$$n \times p$$

dimensions.

For the example:

$$3 \times 3$$

This is called a two-mode matrix because it represents two types of entities: objects and attributes.

4. Dissimilarity matrix (one-mode)

A dissimilarity matrix stores pairwise distances between objects.

For three customers:

	A	B	C
A	0	0.2	0.8
B	0.2	0	0.6
C	0.8	0.6	0

Notice:

- Distance from A to A = 0.
- Distance from A to B = 0.2.
- Distance from B to A = 0.2.

A dissimilarity matrix is symmetric for ordinary symmetric distance measures.

$$d(A, B) = d(B, A)$$

The diagonal contains zeros because an object has no distance from itself.

5. Why are these matrices used?

Data matrix:

Used as the original input for many algorithms.

Dissimilarity matrix:

Used when an algorithm needs pairwise distances between objects.

For example, hierarchical clustering can use a dissimilarity matrix to decide which objects or clusters to merge.

Real-world scenario

Suppose you have 500 stocks and want to group stocks with similar trading behaviour.

First, you create a data matrix containing stock attributes.

Then you calculate pairwise distances.

The resulting dissimilarity matrix tells you which stocks are most similar or dissimilar.

Practice problems

Problem 1: A dataset contains 100 customers and 8 attributes. What is the data matrix dimension?

Answer:

$$100 \times 8$$

Problem 2: A dissimilarity matrix contains three objects. What is its dimension?

Answer:

$$3 \times 3$$

Problem 3: If:

$$d(A, B) = 0.4$$

Calculate similarity.

Answer:

$$1 - 0.4 = 0.6$$

Problem 4: Why are diagonal values zero?

Answer: Because the distance between an object and itself is zero.

Page 26: Similarity for nominal attributes

1. The problem

How do you calculate similarity between two objects if their attributes are categorical?

Consider:

Customer	Department	OS	Division
A	CSE	Android	Dhaka
B	EEE	iOS	Dhaka
C	CSE	Android	Khulna
D	ICT	Android	Dhaka

You cannot use Euclidean distance directly on these categorical values.

Why?

Because the attributes are names, not measurable numbers.

Instead, use simple matching dissimilarity.

2. Formula

$$d(i,j) = \frac{p - m}{p}$$

Where:

- p = Total number of attributes.
- m = Number of matching attributes.
- $p - m$ = Number of mismatching attributes.

In simple words:

$$Dissimilarity = \frac{Number\ of\ mismatches}{Total\ attributes}$$

3. Worked example

Consider:

$$O_1 = (CSE, Android, Dhaka)$$

$$O_2 = (EEE, iOS, Dhaka)$$

There are three attributes.

Compare each attribute:

Attribute	O1	O2	Match?
Programme	CSE	EEE	No
OS	Android	iOS	No
Division	Dhaka	Dhaka	Yes

Matches:

$$m = 1$$

Total attributes:

$$p = 3$$

Therefore:

$$d(O_1, O_2) = \frac{3 - 1}{3} = 2/3$$

$$\boxed{d = 0.667}$$

Similarity:

$$1 - 0.667 = 0.333$$

4. Another example

Consider:

$$O_1 = (CSE, Android, Dhaka)$$
$$O_3 = (CSE, Android, Khulna)$$

Comparison:

- Programme matches.
- OS matches.
- Division does not match.

Therefore:

$$d(O_1, O_3) = \frac{1}{3} = 0.333$$

Similarity:

$$1 - 0.333 = 0.667$$

O1 is more similar to O3 than to O2.

5. Why is it used?

Simple matching is appropriate when categorical attributes are unordered and equally important.

Examples:

- Customer segmentation.
- Product categorization.
- Comparing user profiles.
- Comparing categorical records.

Practice problems

Given:

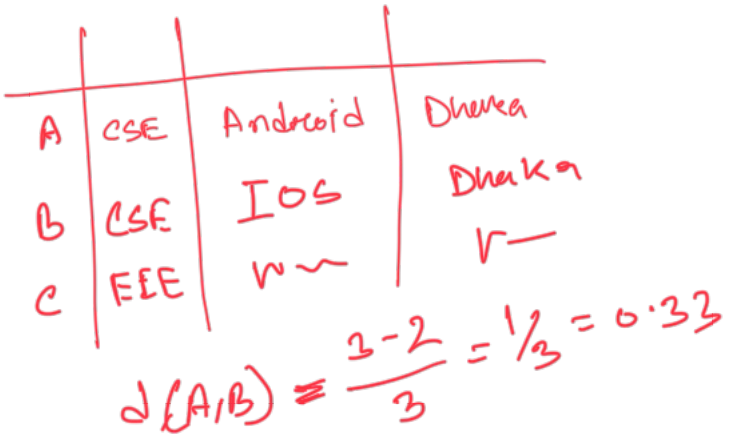
$$A = (CSE, Android, Dhaka)$$

$$B = (CSE, iOS, Dhaka)$$

$$C = (EEE, Windows, Khulna)$$

Calculate:

1. $d(A, B)$.
2. $d(A, C)$.
3. Which pair is more similar?



Solution:

A vs B:

- One mismatch.
- Three attributes.

$$d(A, B) = \frac{1}{3} = 0.333$$

A vs C:

- Three mismatches.

$$d(A, C) = \frac{3}{3} = 1$$

Therefore, A and B are more similar.

Page 27: Binary attributes – Symmetric and asymmetric measures

This is a particularly important mathematical topic.

1. Why do we need a different measure for binary attributes?

Consider two customers.

Customer A:

$$(1, 0, 0, 0, 0)$$

Customer B:

$$(1, 0, 0, 0, 1)$$

Suppose the attributes represent whether customers purchased five products.

A 1 means purchased.

A 0 means not purchased.

They share several zeros, but only some positive purchasing behaviour.

Should the shared zeros make them highly similar?

That depends on the problem.

For asymmetric binary data, shared zeros should not contribute to similarity.

2. The binary contingency table

The lecture introduces four values:

- q
- r
- s
- t

They represent the number of occurrences of each combination.

	Object j = 1	Object j = 0	Total
Object i = 1	q	r	q+r
Object i = 0	s	t	s+t
Total	q+s	r+t	p

Meaning:

Symbol	Meaning
q	Both objects have 1
r	First has 1, second has 0
s	First has 0, second has 1

Symbol	Meaning
t	Both objects have 0

Remember this table. You will use it in examination calculations.

3. Symmetric binary dissimilarity

Formula:

$$d(i,j) = \frac{r+s}{q+r+s+t}$$

Why?

Because both positive and negative mismatches matter.

The denominator includes every attribute.

4. Asymmetric binary dissimilarity

Formula:

$$d(i,j) = \frac{r+s}{q+r+s}$$

Notice that t is removed.

Why?

Because shared zeros are ignored.

5. Jaccard similarity

Formula:

$$Jaccard = \frac{q}{q+r+s}$$

It measures the proportion of shared positive attributes among attributes where at least one object has a positive value.

For asymmetric binary data:

$$Jaccard = 1 - d(i,j)$$

6. Worked example

Suppose:

$$A = (1, 1, 0, 0, 1)$$

$$B = (1, 0, 1, 0, 1)$$

Compare:

Attribute	A	B	Result
1	1	1	q
2	1	0	r
3	0	1	s
4	0	0	t
5	1	1	q

Therefore:

$$q = 2, \quad r = 1, \quad s = 1, \quad t = 1$$

Symmetric dissimilarity:

$$d = \frac{1+1}{2+1+1+1}$$
$$= \frac{2}{5} = 0.4$$

Asymmetric dissimilarity:

$$d = \frac{1+1}{2+1+1}$$
$$= \frac{2}{4} = 0.5$$

Jaccard similarity:

$$J = \frac{2}{2+1+1}$$
$$= \frac{2}{4} = 0.5$$

7. Real-world scenario: Market-basket analysis

Suppose an e-commerce company studies whether customers purchase:

- Laptop.
- Mouse.
- Keyboard.
- Monitor.
- Headphones.

A 1 means purchased, and 0 means not purchased.

If two customers did not purchase a monitor, this does not necessarily mean they have similar shopping interests.

However, if both purchased the same laptop and mouse, that is meaningful.

Therefore, Jaccard similarity is often appropriate.

Practice problems

Given:

$$A = (1, 1, 0, 0, 1, 0)$$

$$B = (1, 0, 0, 1, 1, 0)$$

Calculate:

1. q.
2. r.
3. s.
4. t.
5. Symmetric dissimilarity.
6. Asymmetric dissimilarity.
7. Jaccard similarity.

Solution:

Compare positions:

- Position 1: q.
- Position 2: r.
- Position 3: t.
- Position 4: s.
- Position 5: q.

- Position 6: t.

Therefore:

$$q = 2, \quad r = 1, \quad s = 1, \quad t = 2$$

Symmetric:

$$d = \frac{1+1}{2+1+1+2}$$
$$= \frac{2}{6} = 0.333$$

Asymmetric:

$$d = \frac{1+1}{2+1+1}$$
$$= \frac{2}{4} = 0.5$$

Jaccard:

$$J = \frac{2}{2+1+1} = 0.5$$

Page 28: Worked example – Asymmetric binary attributes

Let's understand the lecture's medical example carefully.

Three patients have these attributes:

- Fever.
- Cough.
- Test-1.
- Test-2.
- Test-3.
- Test-4.

Here:

- 1 = Present or positive.
- 0 = Absent or negative.

The patient data:

$$Rahim = (1,0,1,0,0,0)$$
$$Karim = (1,0,1,0,1,0)$$
$$Salma = (1,1,0,0,0,0)$$

$q=2, r=0, s=1, t=3$

1. Compare Rahim and Karim

Attribute	Rahim	Karim	Result
Fever	1	1	q
Cough	0	0	t
Test-1	1	1	q
Test-2	0	0	t
Test-3	0	1	s
Test-4	0	0	t

Therefore:

$$q = 2, \quad r = 0, \quad s = 1, \quad t = 3$$

Asymmetric dissimilarity:

$$d = \frac{r+s}{q+r+s}$$
$$= \frac{0+1}{2+0+1}$$
$$= \frac{1}{3} = 0.33$$

Jaccard:

$$J = \frac{q}{q+r+s}$$
$$= \frac{2}{3} = 0.67$$

2. Compare Rahim and Salma

Their vectors:

$$Rahim = (1, 0, 1, 0, 0, 0)$$

$$Salma = (1, 1, 0, 0, 0, 0)$$

Comparison:

- q = 1.
- r = 1.
- s = 1.
- t = 3.

Asymmetric dissimilarity:

$$d = \frac{1+1}{1+1+1}$$

$$= \frac{2}{3} = 0.67$$

Jaccard:

$$J = \frac{1}{3} = 0.33$$

3. Compare Karim and Salma

Comparison:

- q = 1.
- r = 2.
- s = 1.
- t = 2.

Asymmetric dissimilarity:

$$d = \frac{2+1}{1+2+1}$$

$$= \frac{3}{4} = 0.75$$

Jaccard:

$$J = \frac{1}{4} = 0.25$$

4. Final comparison

Pair	q	r	s	Asymmetric distance	Jaccard similarity
Rahim–Karim	2	0	1	0.33	0.67
Rahim–Salma	1	1	1	0.67	0.33
Karim–Salma	1	2	1	0.75	0.25

Conclusion:

Rahim and Karim are the most similar pair.

Karim and Salma are the least similar pair.

5. Why is this example important?

It shows why choosing the correct similarity measure matters.

If we treat the data as symmetric, shared negative values contribute to similarity.

In the lecture's example, Rahim and Karim have three shared negatives.

Their symmetric dissimilarity would be:

$$\frac{1}{6} = 0.17$$

But this makes them appear more similar partly because they both lack several symptoms or positive test results.

For this particular problem, asymmetric treatment is more appropriate.

Practice problem

Two patients have:

$$A = (1, 0, 1, 0, 1)$$

$$B = (1, 1, 0, 0, 1)$$

Calculate Jaccard similarity.

Solution:

- q = 2.
- r = 1.
- s = 1.

- t = 1.

$$J = \frac{2}{2+1+1}$$

$$\boxed{J = 0.5}$$

Page 29: Numeric data – Why normalize first?

This is one of the most important concepts for machine learning and data mining.

1. What is normalization?

Normalization is the process of transforming numerical attributes into a comparable scale.

Why do we need it?

Consider two customers:

Attribute	Customer A	Customer B
Age	25	30
Annual income	300,000	500,000

If you calculate Euclidean distance directly:

$$d(A,B) = \sqrt{(25-30)^2 + (300000-500000)^2}$$

$$= \sqrt{25 + 40000000000}$$

$$\approx 200000$$

The income difference dominates the distance.

Age contributes almost nothing.

But what if age is also important in customer segmentation?

The algorithm may fail to recognize the importance of age because the attributes use different numerical scales.

Normalization solves this problem by bringing attributes to comparable scales.

2. Min-max normalization

Min-max normalization transforms values into a specified range.

The most common range is:

$$[0, 1]$$

Formula from the lecture:

$$v' = \frac{v - Min_A}{Max_A - Min_A}$$

For a general target range $[newMin, newMax]$:

$$v' = \frac{v - Min_A}{Max_A - Min_A}(newMax - newMin) + newMin$$

Where:

- v = Original value.
- Min_A = Minimum attribute value.
- Max_A = Maximum attribute value.
- v' = Normalized value.

3. Worked example

Suppose examination scores are:

$$42, 48, 55, 58, 61, 61, 64, 67, 70, 72, 75, 98$$

You want to normalize the score 70 into the range $0, 1$.

Minimum:

$$Min = 42$$

Maximum:

$$Max = 98$$

Original value:

$$v = 70$$

Apply:

$$v' = \frac{70 - 42}{98 - 42}$$

$$= \frac{28}{56}$$

$$\boxed{v' = 0.5}$$

The original value 70 becomes 0.5.

4. What happens to the minimum and maximum?

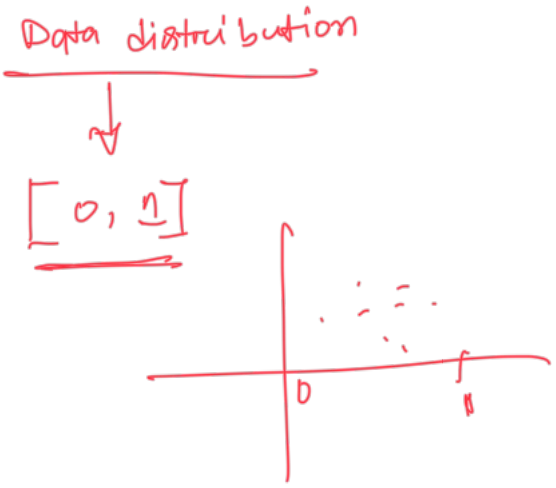
Minimum:

$$\frac{42 - 42}{98 - 42} = 0$$

Maximum:

$$\frac{98 - 42}{98 - 42} = 1$$

All values are mapped into the $0, 1$ range.



5. Z-score normalization

Another important technique is z-score normalization.

Formula:

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

Where:

- v = Original value.
- $\bar{A}$ = Mean.
- σ_A = Standard deviation.

Unlike min-max normalization, z-score normalization does not restrict all values to $0, 1$.

Instead, it expresses how many standard deviations an observation is away from the mean.

6. Worked example

Using the examination data:

Mean:

$$\bar{A} = 64.25$$

Standard deviation:

$$\boxed{\sigma_A = 13.72}$$

$\sigma^2 = \frac{\Sigma x_i^2}{n} - \mu^2$

$\sigma = ?$

Calculate the z-score of 98:

$$z = \frac{98 - 64.25}{13.72}$$

$$= \frac{33.75}{13.72}$$

$$\boxed{z \approx 2.46}$$

Interpretation:

The score 98 is approximately 2.46 standard deviations above the mean.

7. Min-max vs z-score

Property	Min-max	Z-score
Main purpose	Rescale values	Centre and scale values
Typical range	$0, 1$	No fixed range
Uses minimum and maximum	Yes	No
Uses mean and standard deviation	No	Yes
Outlier sensitivity	High	High
Useful for	Bounded scaling	Standardizing numerical features

8. Why is normalization used in machine learning?

Consider KNN.

KNN calculates distances between objects.

If one feature has values between 0 and 1 and another has values between 0 and 1,000,000, the larger-scale feature can dominate.

Normalization helps prevent this.

Common applications:

- KNN.
- K-Means.
- Distance-based clustering.
- Some neural network workflows.

- Similarity calculations.

Not every machine learning algorithm requires normalization. For example, decision trees generally do not depend on numerical distance in the same way.

9. Real-world scenario: Stock-market analysis

Suppose your dataset contains:

Attribute	Value
Stock price	250
Trading volume	500,000
Daily return	2.5
Volatility	0.08

Without scaling, trading volume could dominate a distance calculation.

You should normalize the features before applying distance-based clustering.

Practice problems

Problem 1: Min-max normalize 60 if:

$$Min = 20, \quad Max = 100$$

Solution:

$$v' = \frac{60 - 20}{100 - 20}$$
$$= \frac{40}{80} = 0.5$$

Problem 2: Calculate the z-score if:

$$v = 80, \quad Mean = 60, \quad SD = 10$$

Solution:

$$z = \frac{80 - 60}{10} = 2$$

Interpretation: The value is two standard deviations above the mean.

Problem 3: Why is normalization important before KNN?

Answer: Because KNN relies on distance, and different feature scales can distort the calculated distances.

Page 30: Minkowski distance family

Now we come to one of the most important mathematical topics in the chapter.

You need to understand four measurements:

1. Manhattan distance.
2. Euclidean distance.
3. Minkowski distance.
4. Supremum distance.

All four belong to the Minkowski family.

1. What is distance in data mining?

Suppose two customers have these attributes:

Customer A:

$$A = (2, 7, 3)$$

Customer B:

$$B = (5, 3, 3)$$

We want to measure how different they are.

The difference in each dimension is:

$$|A - B| = |2 - 5, 7 - 3, 3 - 3|$$
$$= (3, 4, 0)$$

Different distance formulas calculate this difference in different ways.

2. General Minkowski formula

$$d(i, j) = \left(\sum_{f=1}^{p} |x_{if} - x_{jf}|^h \right)^{1/h}$$

Where:

- i, j = Two objects.
- p = Number of attributes.
- x_{if} = Attribute f of object i.
- x_{jf} = Attribute f of object j.
- h = Distance parameter.

Changing h changes the distance calculation.

3. Manhattan distance (L1)

When:

$$h = 1$$

The formula becomes:

$$d(i,j) = \sum_{f=1}^{p} |x_{if} - x_{jf}|$$

Using our example:

$$A = (2, 7, 3)$$
$$B = (5, 3, 3)$$

Absolute differences:

$$(3, 4, 0)$$

Therefore:

$$d = 3 + 4 + 0$$

$$\boxed{d = 7}$$

Why is it called Manhattan distance?

Imagine travelling through city streets arranged in a grid.

You cannot move diagonally through buildings.

You must travel along horizontal and vertical streets.

The total street distance resembles Manhattan distance.

When is it used?

- Grid-based movement.
- High-dimensional numerical data.
- Some clustering problems.
- When the sum of absolute differences is appropriate.

4. Euclidean distance (L2)

When:

$$h = 2$$

The formula becomes:

$$d(i,j) = \sqrt{\sum_{f=1}^{p}(x_{if} - x_{jf})^2}$$

Using the example:

$$d = \sqrt{3^2 + 4^2 + 0^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$\boxed{d = 5}$$

What does it represent?

It represents the straight-line distance between two points.

Imagine two locations on a map.

Instead of travelling along city streets, you measure the direct straight-line distance between them.

When is it used?

- KNN.
- K-Means.
- Geometric clustering.
- Spatial analysis.
- Numerical feature comparison.

5. Minkowski distance (L3 example)

When:

$$h = 3$$

$$d = \left(\sum |x_i - x_j|^3 \right)^{1/3}$$

Using our differences:

$$d = (3^3 + 4^3 + 0^3)^{1/3}$$

$$= (27 + 64)^{1/3}$$

$$= 91^{1/3}$$

$$\boxed{d \approx 4.50}$$

6. Supremum distance (L infinity)

When:

$$h \to \infty$$

The distance becomes the largest individual difference.

$$d(i, j) = \max_f |x_{if} - x_{jf}|$$

Using our example:

$$(3, 4, 0)$$

Largest difference:

$$\boxed{d = 4}$$

This is also known as Chebyshev distance.

Why is it used?

When the largest difference in any one dimension is especially important.

Examples:

- Measuring maximum component deviation.
- Checking whether all attributes remain within an acceptable tolerance.
- Certain board-game and grid-distance problems.

7. Compare all four

For:

$$A = (2, 7, 3)$$

$$B = (5, 3, 3)$$

Distance	Formula	Answer
Manhattan	$3 + 4 + 0$	7
Euclidean	$\sqrt{9 + 16}$	5
Minkowski L3	$\sqrt[3]{27 + 64}$	4.50
Supremum	$\max(3, 4, 0)$	4

The lecture highlights:

$$L1 \geq L2 \geq L3 \geq L\infty$$

For the same coordinate differences, increasing h progressively reduces the resulting Minkowski distance towards the maximum individual difference.

8. Properties of a metric

A valid metric distance must satisfy four properties.

Property 1: Non-negativity

$$d(i, j) \geq 0$$

Distance cannot be negative.

Property 2: Identity

$$d(i, i) = 0$$

An object has zero distance from itself.

Property 3: Symmetry

$$d(i, j) = d(j, i)$$

The distance from A to B equals the distance from B to A.

Property 4: Triangle inequality

$$d(i,j) \leq d(i,k) + d(k,j)$$

Direct distance cannot exceed the sum of distances through another point.

9. Real-world scenario

Suppose you want to cluster customers based on:

- Age.
- Monthly income.
- Monthly spending.

You normalize the numerical features and then calculate distances.

Depending on the problem, you may use:

- Euclidean for straight-line numerical distance.
- Manhattan for summed absolute differences.
- Supremum when the largest individual difference is important.

Practice problems

Given:

$$A = (1,2,3)$$
$$B = (4,6,3)$$

Calculate:

1. Manhattan distance.
2. Euclidean distance.
3. Minkowski distance for h = 3.
4. Supremum distance.

Solution:

Absolute differences:

$$(3,4,0)$$

Manhattan:

$$3 + 4 + 0 = \boxed{7}$$

Euclidean:

$$\sqrt{9+16} = \boxed{5}$$

Minkowski L3:

$$(27+64)^{1/3}$$

$$= \boxed{4.50 \text{ approximately}}$$

Supremum:

$$\max(3,4,0) = \boxed{4}$$

Page 31: Worked example – Minkowski distances

The lecture provides this example:

$$x_1 = (2,7,3)$$

$$x_2 = (5,3,3)$$

First calculate the absolute differences:

$$\boxed{|x_1 - x_2| = (3,4,0)}$$

Now calculate each distance.

1. Manhattan distance

$$L1 = 3+4+0$$

$$\boxed{L1 = 7}$$

2. Euclidean distance

$$L2 = \sqrt{3^2+4^2+0^2}$$

$$= \sqrt{25}$$

$$\boxed{L2 = 5}$$

3. Minkowski distance for h = 3

$$L3 = (3^3+4^3+0^3)^{1/3}$$

$$= 91^{1/3}$$

$$\boxed{L3 \approx 4.50}$$

4. Supremum distance

$$L\infty = \max(3, 4, 0)$$

$$\boxed{L\infty = 4}$$

5. Important mathematical observation

$$7 > 5 > 4.50 > 4$$

Why does the distance decrease?

Because higher powers give relatively more influence to the largest individual difference, and the final root changes the scale of the result.

As h approaches infinity, the distance approaches the maximum coordinate difference.

Practice problem

Given:

$$A = (3, 5, 7)$$

$$B = (1, 2, 7)$$

Calculate all four distances.

Answer:

Absolute difference:

$$(2, 3, 0)$$

- Manhattan = 5.
- Euclidean = $\sqrt{13} \approx 3.61$.
- Minkowski L3 = $\sqrt[3]{35} \approx 3.27$.
- Supremum = 3.

Page 32: Ordinal attributes – Measuring distance

We already learned that ordinal attributes have an order but do not necessarily have equal intervals.

Now the question is:

How can we calculate distances between ordinal attributes?

For example:

$$Poor < Fair < Good < Excellent$$

We cannot directly subtract these words.

The lecture gives a three-step solution.

Step 1: Replace categories with ranks

Assign ranks:

Category	Rank
Poor	1
Fair	2
Good	3
Excellent	4

Here:

$$M = 4$$

M represents the number of possible ordered categories.

Step 2: Normalize the ranks

Formula:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Where:

- r_{if} = Rank of the object's attribute.
- M_f = Number of categories.
- z_{if} = Normalized rank.

For four categories:

Poor:

$$\frac{1 - 1}{4 - 1} = 0$$

Fair:

$$\frac{2-1}{3} = 0.33$$

Good:

$$\frac{3-1}{3} = 0.67$$

Excellent:

$$\frac{4-1}{3} = 1$$

Normalized values: (0, 0.33, 0.67, 1)

Category	Rank	Normalized value
Poor	1	0
Fair	2	0.33
Good	3	0.67
Excellent	4	1

Step 3: Apply a numeric distance measure

Now you can calculate differences.

Suppose:

$$A = Excellent = 1$$

$$B = Fair = 0.33$$

Using absolute difference:

$$d(A, B) = |1 - 0.33|$$

$$\boxed{d = 0.67}$$

Another example:

$$A = Excellent = 1$$

$$C = Good = 0.67$$

$$d(A, C) = |1 - 0.67|$$

$$\boxed{d = 0.33}$$

Why is it used?

It allows us to calculate proximity between ranked categorical data.

Applications:

- Customer satisfaction.
- Product quality.
- Employee performance.
- Credit-risk ratings.
- Academic performance.

Real-world scenario

Suppose a bank classifies customers by credit risk:

- Very Low.
- Low.
- Medium.
- High.
- Very High.

You want to calculate the similarity between two customers based on their risk ratings.

By converting ranks into normalized values, you can use a numeric distance calculation.

Important limitation

The rank normalization method treats adjacent categories as equally spaced after conversion.

That is a modelling assumption.

The original ordinal data does not guarantee equal real-world differences between categories.

Practice problem

Given:

$$Poor < Fair < Good < Excellent$$

Customer A = Good.

Customer B = Poor.

Calculate normalized values and their distance.

Solution:

Good:

$$z_A = \frac{3-1}{4-1} = \frac{2}{3} = 0.67$$

Poor:

$$z_B = \frac{1-1}{3} = 0$$

Distance:

$$|0.67 - 0| = \boxed{0.67}$$

Page 33: Mixed attribute types

This is one of the most practically useful topics in the chapter.

1. What is mixed data?

In real-world datasets, not every attribute has the same type.

Consider a customer database:

Customer	Department	Satisfaction	Age	Purchased
A	CSE	Good	25	Yes
B	EEE	Fair	30	No
C	CSE	Excellent	28	Yes

Here you have:

- Department: Nominal.
- Satisfaction: Ordinal.
- Age: Numeric.
- Purchased: Binary.

How can you calculate similarity between customers when attributes have different data types?

You cannot blindly apply Euclidean distance to the entire dataset.

The lecture introduces a mixed-attribute dissimilarity measure.

2. General formula

$$d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$$

This is a weighted average of attribute-level dissimilarities.

Where:

- p = Total number of attributes.
- $d_{ij}^{(f)}$ = Dissimilarity for attribute f.
- $\delta_{ij}^{(f)}$ = Indicator determining whether the attribute should be included.

3. What is delta?

This is important.

$$\delta_{ij}^{(f)} = 0$$

Means the attribute is excluded from the comparison.

$$\delta_{ij}^{(f)} = 1$$

Means the attribute is included.

According to the lecture, an attribute is excluded when:

- Either object has a missing value.
- Both values are 0 for an asymmetric binary attribute.

This prevents inappropriate comparisons.

4. How to calculate different attribute types?

The lecture gives separate rules.

Numeric attribute

$$d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{Max_f - Min_f}$$

This is normalized absolute difference.

Nominal attribute

$$d_{ij}^{(f)} = \begin{cases} 0 & \text{if equal} \\ 1 & \text{if different} \end{cases}$$

Binary attribute

For the applicable binary convention, use 0 for a match and 1 for a mismatch, excluding shared negative matches for asymmetric binary attributes.

Ordinal attribute

First normalize the ranks:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Then treat the resulting values as numeric.

5. Worked example with three attribute types

Suppose we compare two students using:

- Programme (Nominal).
- Satisfaction (Ordinal).
- Study hours (Numeric).

Student A:

$$A = (CSE, Good, 5)$$

Student B:

$$B = (EEE, Fair, 8)$$

Assume the satisfaction order is:

$$Poor < Fair < Good < Excellent$$

And study hours in the dataset range from 0 to 10.

Step 1: Nominal distance

CSE vs EEE.

They are different.

$$d_{nominal} = 1$$

Step 2: Ordinal distance

Good:

$$z_A = \frac{3-1}{4-1} = 0.67$$

Fair:

$$z_B = \frac{2-1}{4-1} = 0.33$$

Distance:

$$d_{ordinal} = |0.67 - 0.33| = 0.34$$

Using exact thirds:

$$d_{ordinal} = \frac{1}{3}$$

Step 3: Numeric distance

Study hours:

$$d_{numeric} = \frac{|5-8|}{10-0}$$

$$= \frac{3}{10} = 0.3$$

Step 4: Calculate overall distance

Assuming all three attributes are present and equally weighted:

$$d(A,B) = \frac{1 + 0.333 + 0.3}{3}$$

$$= \frac{1.633}{3}$$

$$\boxed{d(A,B) \approx 0.544}$$

6. Why is it used?

Most business datasets are mixed.

For example, a banking dataset might contain:

- Age.

- Salary.
- Account type.
- Marital status.
- Credit risk.
- Purchased product.
- Number of transactions.

Mixed-attribute distance allows you to compare such records using appropriate calculations for each attribute.

7. Real-world scenario

Suppose you are developing a customer segmentation system.

You have 50,000 customers.

Each customer contains numerical, categorical, ordinal and binary attributes.

You want to group similar customers.

A mixed dissimilarity measure allows you to combine the different attribute types rather than discarding useful information.

Practice problem

Two objects contain:

Attribute	A	B	Type
Department	CSE	CSE	Nominal
Rating	Good	Fair	Ordinal
Age	20	30	Numeric

Assume rating categories are:

Poor, Fair, Good, Excellent.

Assume age ranges from 10 to 50.

Calculate the overall dissimilarity using equal weights.

Solution:

Nominal:

$$d_1 = 0$$

Ordinal:

$$Good = 0.67, \quad Fair = 0.33$$

$$d_2 = 0.33$$

Numeric:

$$d_3 = \frac{|20 - 30|}{50 - 10}$$

$$= \frac{10}{40} = 0.25$$

Overall:

$$d = \frac{0 + 0.33 + 0.25}{3}$$

$$\boxed{d \approx 0.193}$$

Page 34: Worked example – Mixed types

The lecture gives a comparison table showing dissimilarities for six pairs of objects.

Pair	Nominal	Ordinal	Numeric	Overall distance
O1, O2	1	0.667	0.548	0.738
O1, O3	0	0.333	0.452	0.262
O1, O4	1	0	0.405	0.468
O2, O3	1	0.333	1.000	0.778
O2, O4	1	0.667	0.143	0.603
O3, O4	1	0.333	0.857	0.730

1. How do we interpret this table?

Each pair has three attribute-level dissimilarities.

The overall distance combines these dissimilarities.

For O1 and O2:

$$d(O1, O2) = 0.738$$

For O1 and O3:

$$d(O1, O3) = 0.262$$

Smaller distance means greater similarity.

Therefore, O1 and O3 are more similar.

For O2 and O3:

$$d(O2, O3) = 0.778$$

This is the highest distance in the table.

Therefore, O2 and O3 are the least similar.

2. Why is this example important?

It demonstrates that different attribute types can be combined into one final proximity score.

You do not have to calculate a completely separate similarity system for every attribute type and abandon the results.

Instead, you calculate each attribute's contribution using the appropriate rule and combine them.

3. Real-world scenario

Imagine comparing four customers using:

- Customer category.
- Satisfaction rating.
- Annual spending.

The overall distance allows you to identify which customers have the most similar profiles.

This is useful before clustering.

Practice problems

Based on the lecture table:

1. Which pair is most similar?
2. Which pair is least similar?
3. What does a distance of 0.262 mean?
4. Does a smaller distance indicate more similarity?

Answers:

1. O1 and O3.
2. O2 and O3.
3. The pair has relatively low dissimilarity.
4. Yes.

Page 35: Cosine similarity and Tanimoto coefficient

This is another very important topic, especially for text mining, document similarity and market-basket analysis.

1. The problem with ordinary distance

Imagine two documents discussing artificial intelligence.

Document A is short:

"Artificial intelligence improves healthcare."

Document B is long:

"Artificial intelligence improves healthcare through machine learning, deep learning, computer vision, natural language processing and predictive analytics."

Both discuss similar topics.

However, the second document contains many more words.

A raw Euclidean distance between their word-count vectors may be affected by document length.

We need a measurement that can focus on the direction or pattern of the vectors rather than just their magnitude.

This is where cosine similarity becomes useful.

2. What is cosine similarity?

Cosine similarity measures the cosine of the angle between two vectors.

Formula:

$$sim(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$$

Where:

- $x \cdot y$ = Dot product.
- $\|x\|$ = Magnitude of vector x.
- $\|y\|$ = Magnitude of vector y.

The result typically ranges from -1 to 1 for general real-valued vectors.

For nonnegative document-frequency vectors, it ranges from 0 to 1.

3. Understanding using an example

Consider:

$$x = (1, 2)$$

$$y = (2, 4)$$

The vectors point in exactly the same direction.

Dot product:

$$x \cdot y = (1 \times 2) + (2 \times 4)$$

$$= 10$$

Magnitude of x:

$$\|x\| = \sqrt{1^2 + 2^2}$$

$$= \sqrt{5}$$

Magnitude of y:

$$\|y\| = \sqrt{2^2 + 4^2}$$

$$= \sqrt{20}$$

Cosine similarity:

$$= \frac{10}{\sqrt{5}\sqrt{20}}$$

$$= \frac{10}{10}$$

1

The vectors have perfect directional similarity.

4. How to interpret cosine similarity?

Cosine similarity	Interpretation
1	Same direction
0.8	Strong directional similarity
0.5	Moderate directional similarity
0	Perpendicular vectors

For nonnegative document vectors, the similarity is generally between 0 and 1.

5. Why is cosine similarity used?

It is particularly useful for sparse, high-dimensional vectors.

Examples:

- Document similarity.
- Search engines.
- Recommendation systems.
- Text classification.
- Information retrieval.

6. What is a sparse vector?

A sparse vector is a vector containing many zeros.

Suppose a document vocabulary contains 10,000 words.

One document may contain only 100 of them.

Its vector representation will contain many zeros.

This is common in text mining.

Cosine similarity can compare these high-dimensional vectors without treating shared zero positions as evidence of similarity.

7. Tanimoto coefficient

The lecture also introduces the Tanimoto coefficient.

Formula:

$$T(x,y) = \frac{x \cdot y}{x \cdot x + y \cdot y - x \cdot y}$$

It is closely related to the Jaccard coefficient for binary vectors.

For binary vectors:

$$Tanimoto = Jaccard$$

8. Worked example from the lecture

Given:

$$x = (3,0,2,1,0,4)$$
$$y = (1,0,3,0,2,2)$$

Step 1: Calculate the dot product.

$$x \cdot y$$
$$= (3 \times 1) + (0 \times 0) + (2 \times 3)$$
$$+(1 \times 0) + (0 \times 2) + (4 \times 2)$$
$$= 3 + 0 + 6 + 0 + 0 + 8$$
$$\boxed{17}$$

Step 2: Calculate magnitude of x.

$$\|x\| = \sqrt{3^2 + 0^2 + 2^2 + 1^2 + 0^2 + 4^2}$$
$$= \sqrt{9 + 4 + 1 + 16}$$
$$= \sqrt{30}$$

Step 3: Calculate magnitude of y.

$$\|y\| = \sqrt{1^2 + 0^2 + 3^2 + 0^2 + 2^2 + 2^2}$$
$$= \sqrt{1 + 9 + 4 + 4}$$
$$= \sqrt{18}$$

Step 4: Cosine similarity.

$$sim(x,y) = \frac{17}{\sqrt{30}\sqrt{18}}$$

$$\boxed{sim \approx 0.73}$$

9. Binary Tanimoto example

Consider:

$$x = (1,0,1,1,0,1)$$

$$y = (1,0,1,0,1,1)$$

Shared positive values:

$$q = 3$$

Number of positive values in x:

$$x \cdot x = 4$$

Number of positive values in y:

$$y \cdot y = 4$$

Dot product:

$$x \cdot y = 3$$

Apply:

$$T = \frac{3}{4+4-3}$$

$$= \frac{3}{5}$$

$$\boxed{T = 0.60}$$

10. Important limitation

The lecture warns that cosine similarity is not itself a metric distance.

In particular:

$$1 - \cos(x,y)$$

can violate the triangle inequality.

This matters when an algorithm requires a true metric distance.

Real-world scenario

Suppose a search engine receives a query:

"Machine learning in healthcare"

It converts the query and documents into vectors.

Cosine similarity can rank documents based on how closely their vector directions match the query.

Another example is comparing product baskets without allowing the absence of thousands of unrelated products to dominate similarity.

Practice problems

Problem 1: Calculate cosine similarity:

$$x = (1, 0, 1)$$

$$y = (1, 1, 0)$$

Solution:

Dot product:

$$1 + 0 + 0 = 1$$

Magnitudes:

$$\|x\| = \sqrt{2}$$

$$\|y\| = \sqrt{2}$$

Therefore:

$$sim = \frac{1}{2}$$

$$\boxed{0.5}$$

Problem 2: Calculate Tanimoto similarity:

$$A = (1, 1, 0, 0)$$

$$B = (1, 0, 1, 0)$$

Shared positive attributes:

$$q = 1$$

Each vector has two positive values.

$$T = \frac{1}{2 + 2 - 1}$$

$$\boxed{T = 0.333}$$

Page 36: Choosing a proximity measure

This page combines all the proximity concepts into one decision table.

The main lesson is:

Do not use the same distance formula for every dataset.

Choose the measure based on the attribute type and the problem.

1. Complete decision table

Data type	Appropriate measure	Important point
Nominal	Simple matching	Count mismatches
Symmetric binary	Simple matching distance	Both 0 and 1 matter
Asymmetric binary	Jaccard	Ignore shared zeros
Numeric	Manhattan, Euclidean, Minkowski, Supremum	Normalize first when scales differ
Ordinal	Rank → Normalize → Numeric distance	Preserve ranking
Mixed	Delta-weighted dissimilarity	Handle missing values and different types
Sparse vectors	Cosine	Focus on vector direction
Binary sparse vectors	Tanimoto / Jaccard	Focus on shared positive values

2. How to select the right measure in an examination

Imagine the examiner gives you a problem.

First, identify the data type.

Scenario A:

Two customers have:

- Department.
- Operating system.
- Division.

These are nominal attributes.

Use simple matching.

Scenario B:

Two patients have:

- Fever.
- Cough.
- Disease-positive tests.

Positive outcomes matter more than shared negatives.

Use asymmetric binary dissimilarity or Jaccard similarity.

Scenario C:

Two students have:

- Age.
- Weight.
- Height.

These are numerical attributes.

Use Minkowski-family distances.

Normalize first if the scales differ.

Scenario D:

Two customers have:

- Satisfaction level.

- Credit-risk category.

These are ordinal attributes.

Convert ranks into normalized values first.

Scenario E:

Two customers have:

- Age.
- Salary.
- Department.
- Customer category.
- Purchase status.

These are mixed attributes.

Use mixed-attribute dissimilarity.

Scenario F:

Two documents contain thousands of possible words.

The vectors contain mostly zeros.

Use cosine similarity.

3. Why does selecting the wrong measure cause problems?

Imagine comparing two customers using purchase data:

Customer A:

$$(1, 0, 0, 0, 0)$$

Customer B:

$$(1, 0, 0, 0, 1)$$

If you use simple matching, shared zeros contribute to similarity.

But if you are trying to understand purchase preferences, the shared positive purchases are more informative.

Jaccard is more appropriate.

A wrong proximity measure can lead to incorrect clusters, misleading recommendations and poor machine learning results.

Practice problems

Select the most appropriate measure.

1. Comparing blood groups.
2. Comparing two shopping baskets.
3. Comparing customer ages and incomes.
4. Comparing customer satisfaction ratings.
5. Comparing customer profiles containing numerical and categorical features.
6. Comparing documents using word-frequency vectors.

Answers:

1. Simple matching.
2. Jaccard.
3. Numeric distance after appropriate scaling.
4. Rank normalization followed by numeric distance.
5. Mixed-attribute dissimilarity.
6. Cosine similarity.

Part 5: Chapter summary and examination preparation

Page 38: Take-home messages

Let's connect the entire chapter.

Topic	What you learned	Why it matters
Data objects	Records representing entities	Identifies what you are analysing
Attributes	Characteristics of objects	Determines data representation
Nominal	Unordered categories	Prevents meaningless numerical calculations

Topic	What you learned	Why it matters
Binary	Two possible states	Helps represent yes/no information
Ordinal	Ordered categories	Allows meaningful ranking
Interval	Equal intervals, no true zero	Supports differences
Ratio	Equal intervals, true zero	Supports meaningful ratios
Mean, median, mode	Central tendency	Describes typical values
Range, IQR, variance	Dispersion	Measures data spread
Boxplot	Five-number summary	Detects potential outliers
Histogram	Frequency distribution	Understands data shape
Scatter plot	Relationship between two variables	Identifies correlation
Visualization families	Multidimensional displays	Understands complex datasets
Similarity	How alike objects are	Supports clustering
Dissimilarity	How different objects are	Supports distance-based algorithms
Normalization	Comparable numerical scales	Prevents scale domination
Minkowski	Numeric distance family	Measures numerical differences
Jaccard	Positive binary overlap	Compares purchasing behaviour
Cosine	Vector direction similarity	Compares documents
Mixed distance	Combines attribute types	Supports real-world datasets

The most important formulas to memorize

Central tendency

Mean:

$$\bar{x} = \frac{\sum x_i}{N}$$

Weighted mean:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Median position for odd N:

$$\frac{N+1}{2}$$

Midrange:

$$\frac{Min + Max}{2}$$

Dispersion

Range:

$$Max - Min$$

IQR:

$$Q3 - Q1$$

Outlier fences:

$$Q1 - 1.5(IQR)$$

$$Q3 + 1.5(IQR)$$

Population variance:

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Standard deviation:

$$\sigma = \sqrt{\sigma^2}$$

Normalization

Min-max:

$$v' = \frac{v - Min}{Max - Min}$$

Z-score:

$$z = \frac{v - \bar{x}}{\sigma}$$

Proximity

Nominal dissimilarity:

$$d = \frac{p - m}{p}$$

Jaccard similarity:

$$J = \frac{q}{q + r + s}$$

Manhattan:

$$L1 = \sum |x_i - y_i|$$

Euclidean:

$$L2 = \sqrt{\sum (x_i - y_i)^2}$$

Supremum:

$$L\infty = \max |x_i - y_i|$$

Cosine similarity:

$$\frac{x \cdot y}{\|x\| \|y\|}$$

Tanimoto:

$$\frac{x \cdot y}{x \cdot x + y \cdot y - x \cdot y}$$

Page 39: Chapter examination practice

These questions are based on the final understanding-check slide of your lecture. I've expanded them with explanations and additional practice.

Question 1: Is a Likert rating of 1–5 interval-scaled? Defend your answer.

Answer: No, not automatically.

A Likert rating is generally treated as ordinal because the values indicate an order.

For example:

Rating	Meaning
1	Strongly disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly agree

We know that:

$$1 < 2 < 3 < 4 < 5$$

However, we cannot guarantee that the difference between 1 and 2 is exactly equal to the difference between 4 and 5.

Therefore, the basic measurement-scale classification is ordinal.

Some statistical analyses treat multi-item Likert scales as approximately interval-scaled under additional assumptions, but that is a separate modelling decision.

Question 2: Find the five-number summary and identify outliers.

Given:

$$8, 12, 15, 15, 18, 21, 24, 27, 30, 55$$

Using the median-of-halves convention:

Step 1: Minimum

$$Min = 8$$

Step 2: Median

There are 10 observations.

$$Median = \frac{18 + 21}{2} = 19.5$$

Step 3: Q1

Lower half:

$$8, 12, 15, 15, 18$$

$$Q1 = 15$$

Step 4: Q3

Upper half:

$$21, 24, 27, 30, 55$$

$$Q3 = 27$$

Step 5: Maximum

$$Max = 55$$

Five-number summary:

$$\boxed{8, 15, 19.5, 27, 55}$$

Step 6: IQR

$$IQR = 27 - 15 = 12$$

Step 7: Outlier fences

Lower:

$$15 - 1.5(12) = -3$$

Upper:

$$27 + 1.5(12) = 45$$

Since:

$$55 > 45$$

55 is a suspected outlier.

(Final answer)

$$(8, 15, 19.5, 27, 55)$$

55 is a suspected outlier.

Question 3: Calculate Manhattan, Euclidean and supremum distances.

Given:

$$A = (1, 4, 6, 2)$$

$$B = (3, 1, 6, 7)$$

Step 1: Calculate absolute differences.

$$|A - B| = (2, 3, 0, 5)$$

Manhattan:

$$2 + 3 + 0 + 5 = \boxed{10}$$

Euclidean:

$$\sqrt{2^2 + 3^2 + 0^2 + 5^2}$$

$$= \sqrt{4 + 9 + 25}$$

$$\boxed{\sqrt{38} \approx 6.16}$$

Supremum:

$$\max(2, 3, 0, 5)$$

$$\boxed{5}$$

Question 4: Why is Jaccard appropriate for market-basket data?

Suppose two customers do not purchase 1,000 products.

Should those 1,000 shared non-purchases make the customers highly similar?

No.

The more meaningful information is which products they actually purchased.

Jaccard similarity measures shared positive purchases:

$$J = \frac{q}{q + r + s}$$

It ignores shared zeros.

Therefore, it is appropriate for asymmetric binary purchasing data.

Question 5: Why can Euclidean distance mislead when comparing a long and short document on the same topic?

A longer document can have larger word-frequency values simply because it contains more words.

Euclidean distance is sensitive to the magnitude of vectors.

Therefore, it may suggest that two documents are dissimilar even when their word distributions point in similar directions.

Cosine similarity addresses this by comparing vector directions rather than raw magnitudes.

Final self-assessment: Can you solve these without looking at the notes?

This is a comprehensive practice set covering the whole chapter.

Chapter 2 knowledge test (/10 answered)

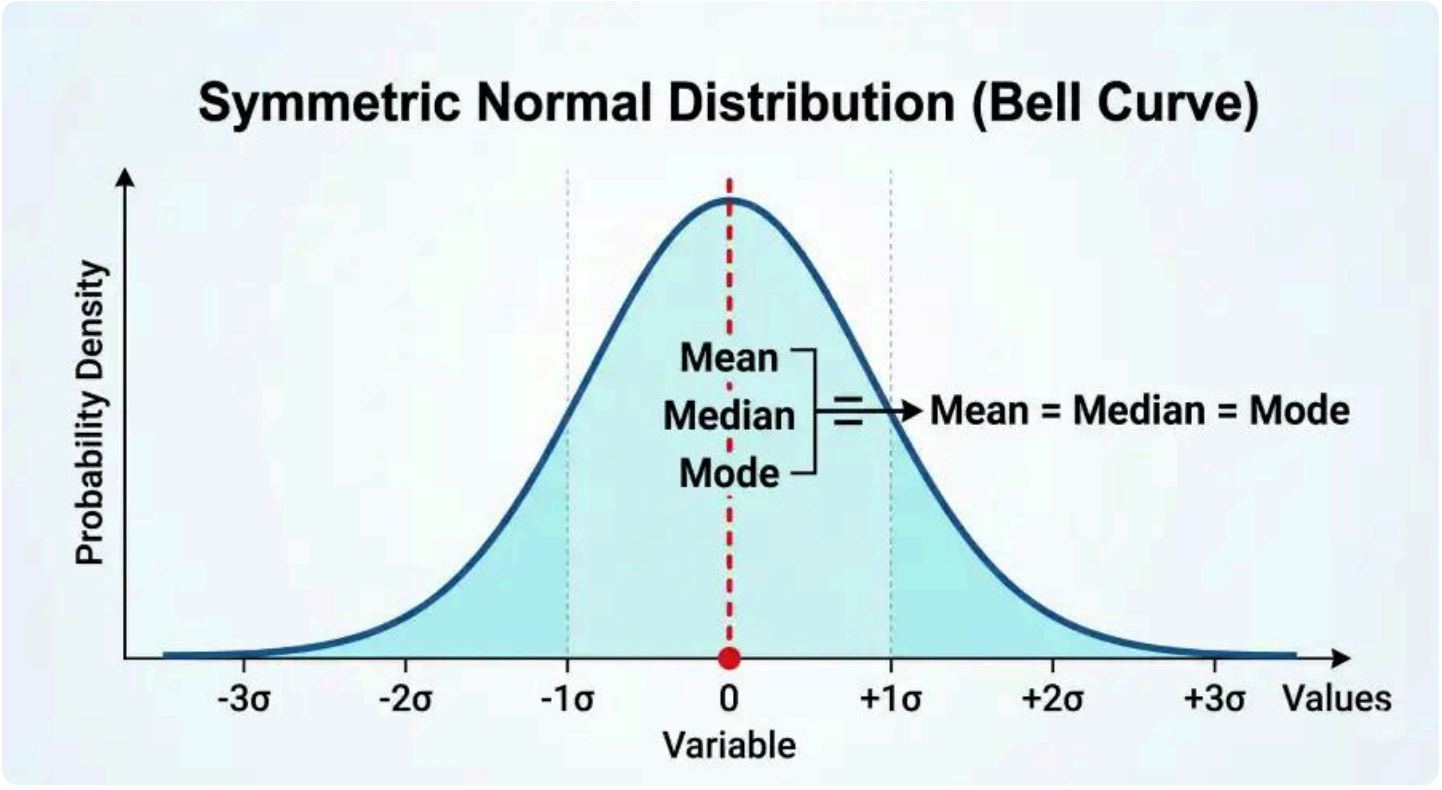
How I recommend studying this chapter

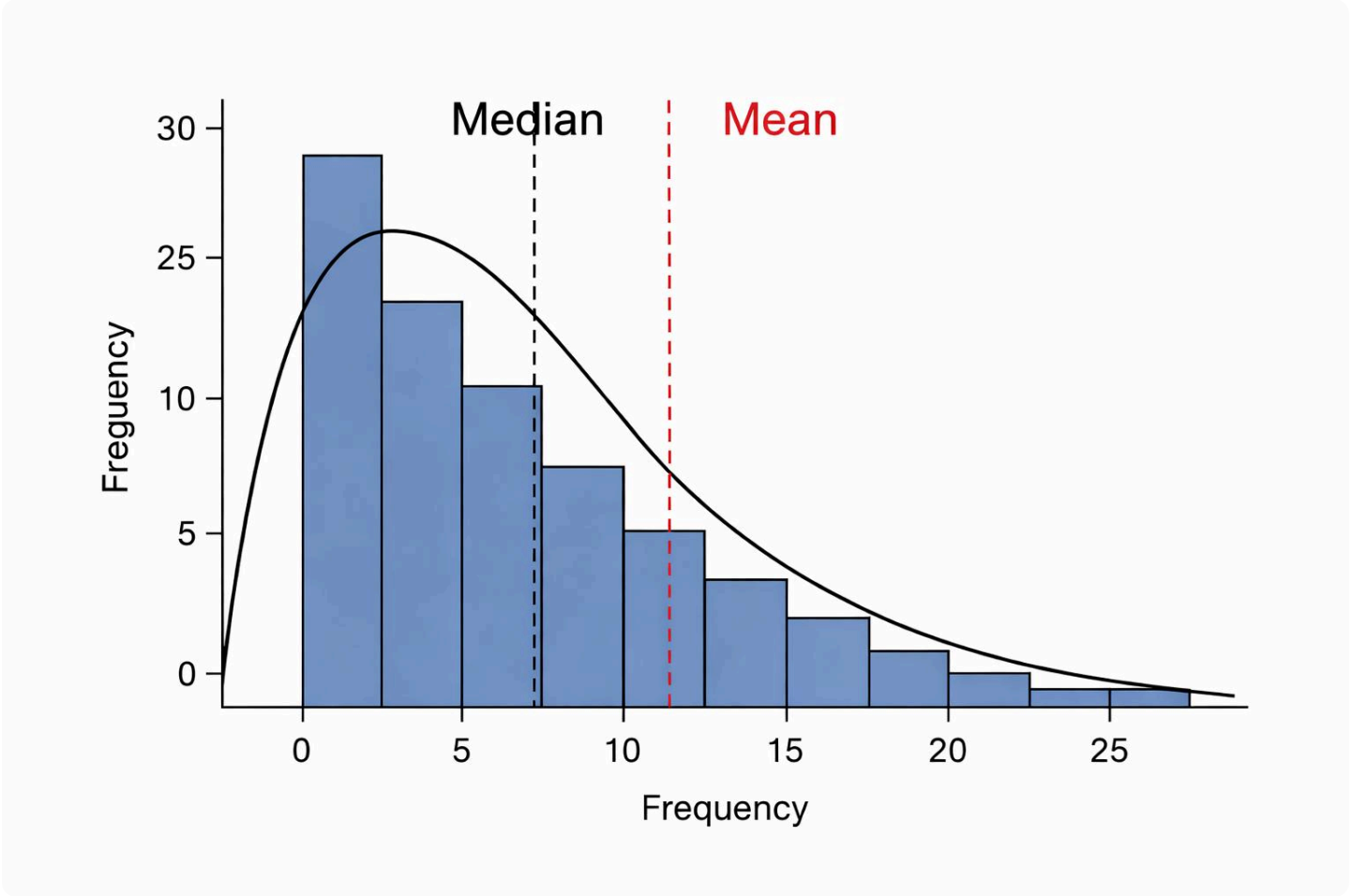
Don't memorize all the formulas at once. Learn them in this order:

1. Attribute types – understand the data before calculations.
2. Mean, median and mode – practise central tendency.
3. Quartiles, IQR and boxplots – practise outlier detection.
4. Variance and standard deviation – practise numerical calculations.
5. Visualization – learn which graph solves which problem.
6. Similarity and dissimilarity – practise the different data types.
7. Minkowski, Jaccard and cosine – master the distance calculations.

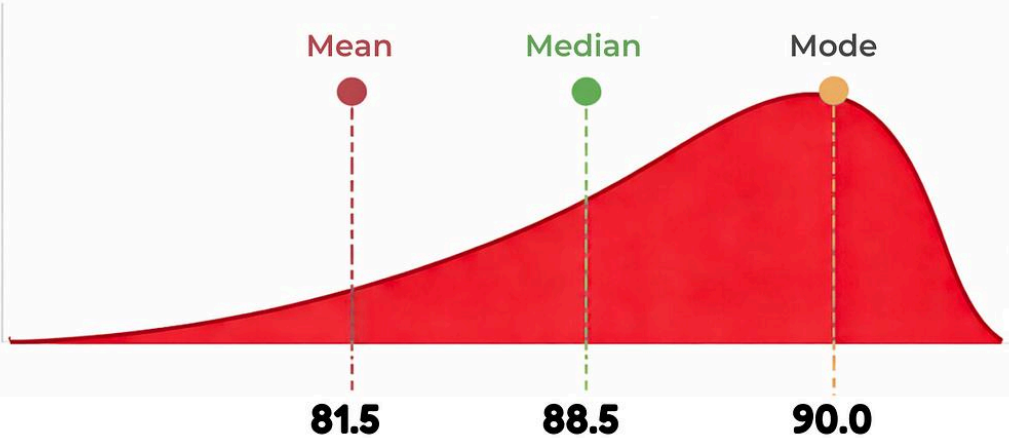
My biggest recommendation: Pay particular attention to normalization, quartile calculations, binary similarity and Minkowski distance. These topics connect multiple concepts and are especially useful in practical data mining.

The explanations above follow your uploaded Chapter 2 lecture, with additional teaching examples and practice exercises. filecite turn0file0

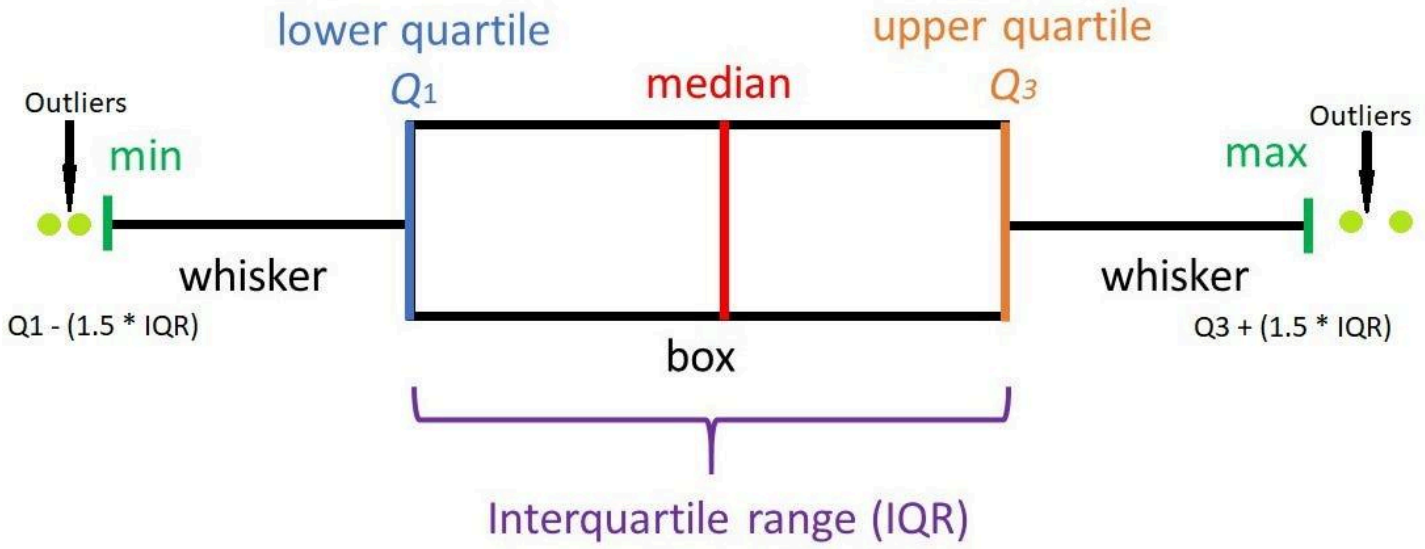

Symmetric Normal Distribution (Bell Curve)
Probability Density
Mean
Median
Mode
=
Mean = Median = Mode
-3σ
-2σ
-1σ
0
+1σ
+2σ
+3σ
Values
Variable

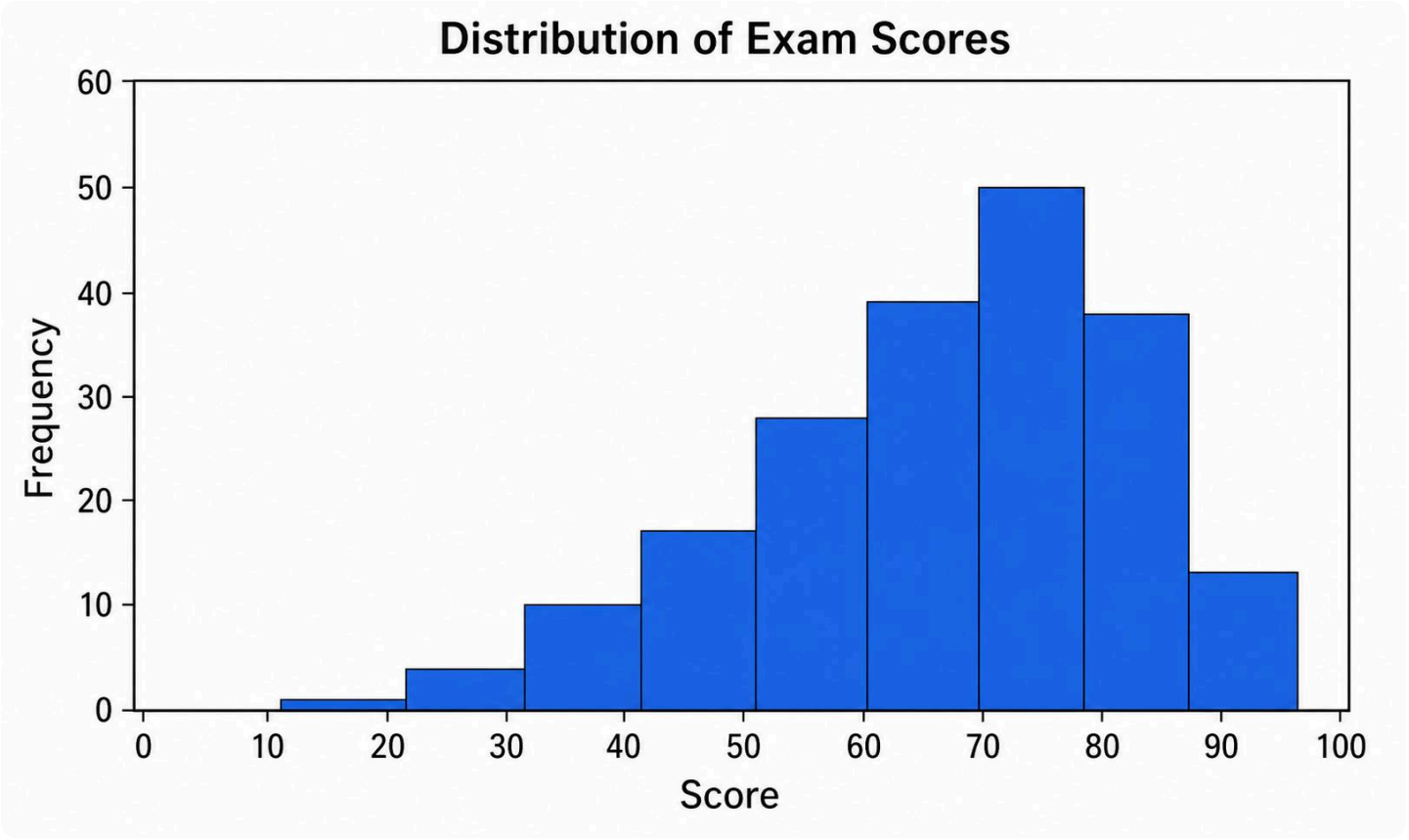

Median
Mean
30
25
10
5
0
Freguency
0
5
10
15
20
25
Frequency

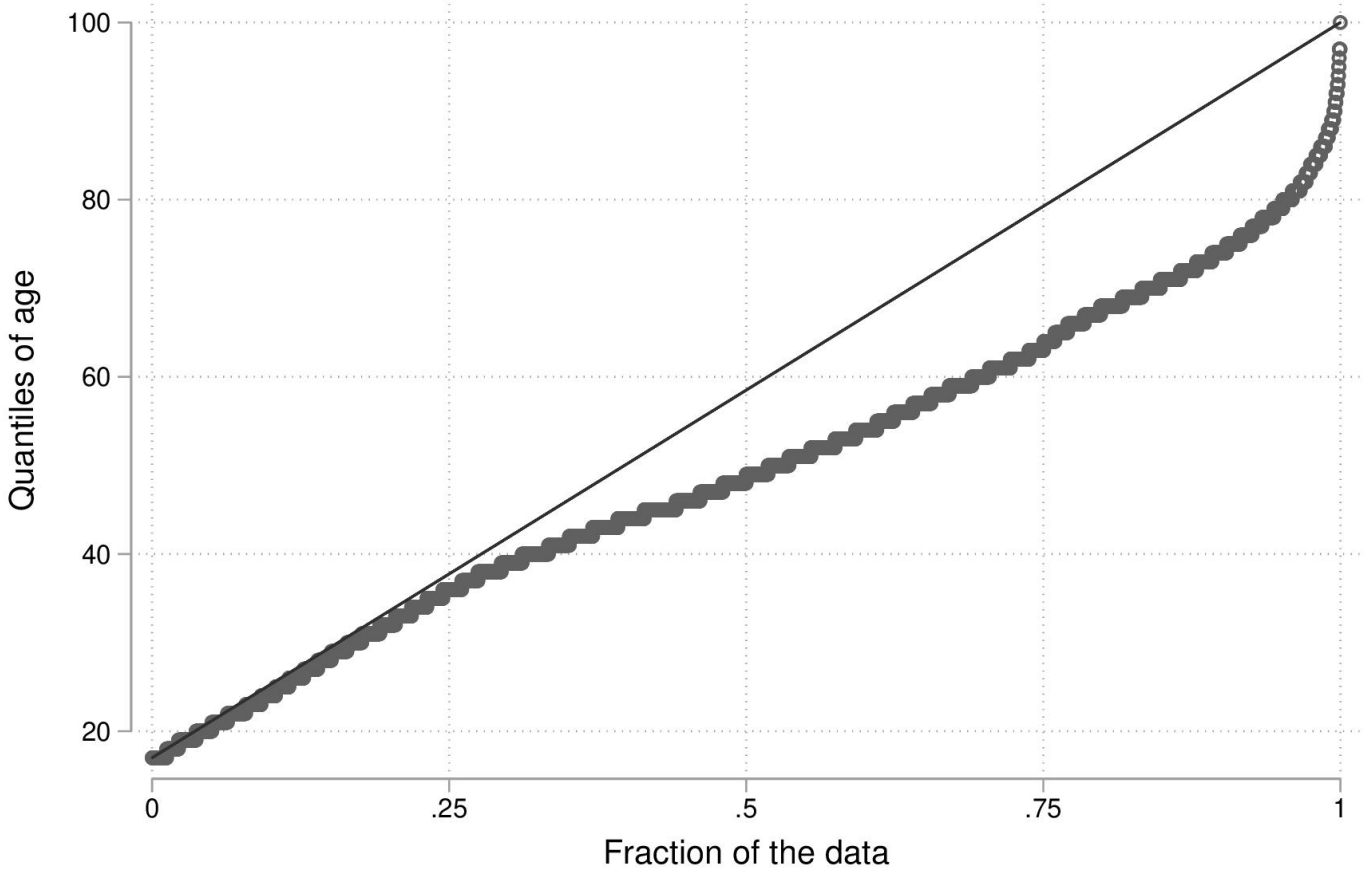
NEGATIVE SKEWNESS (LEFT SKEW)

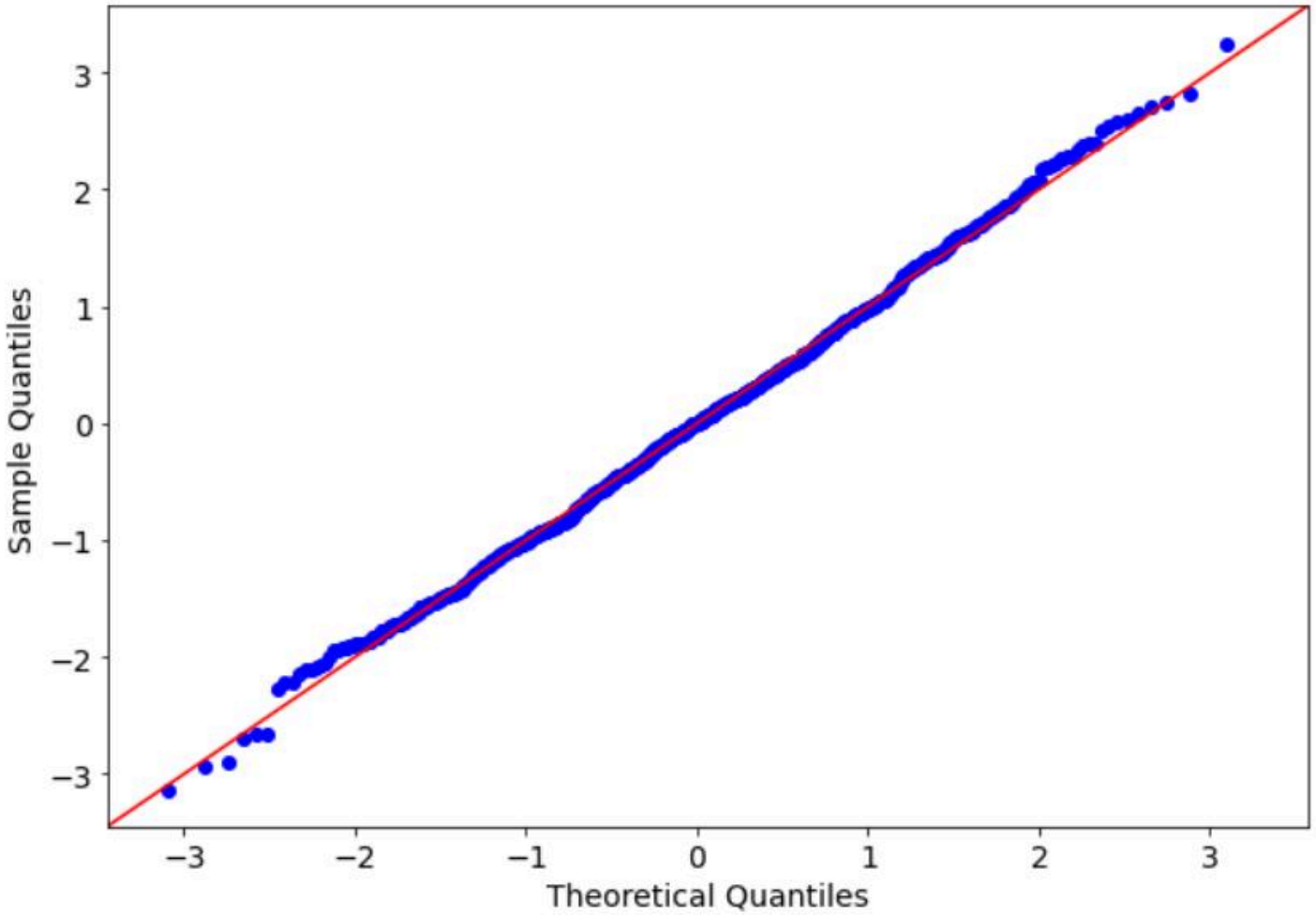


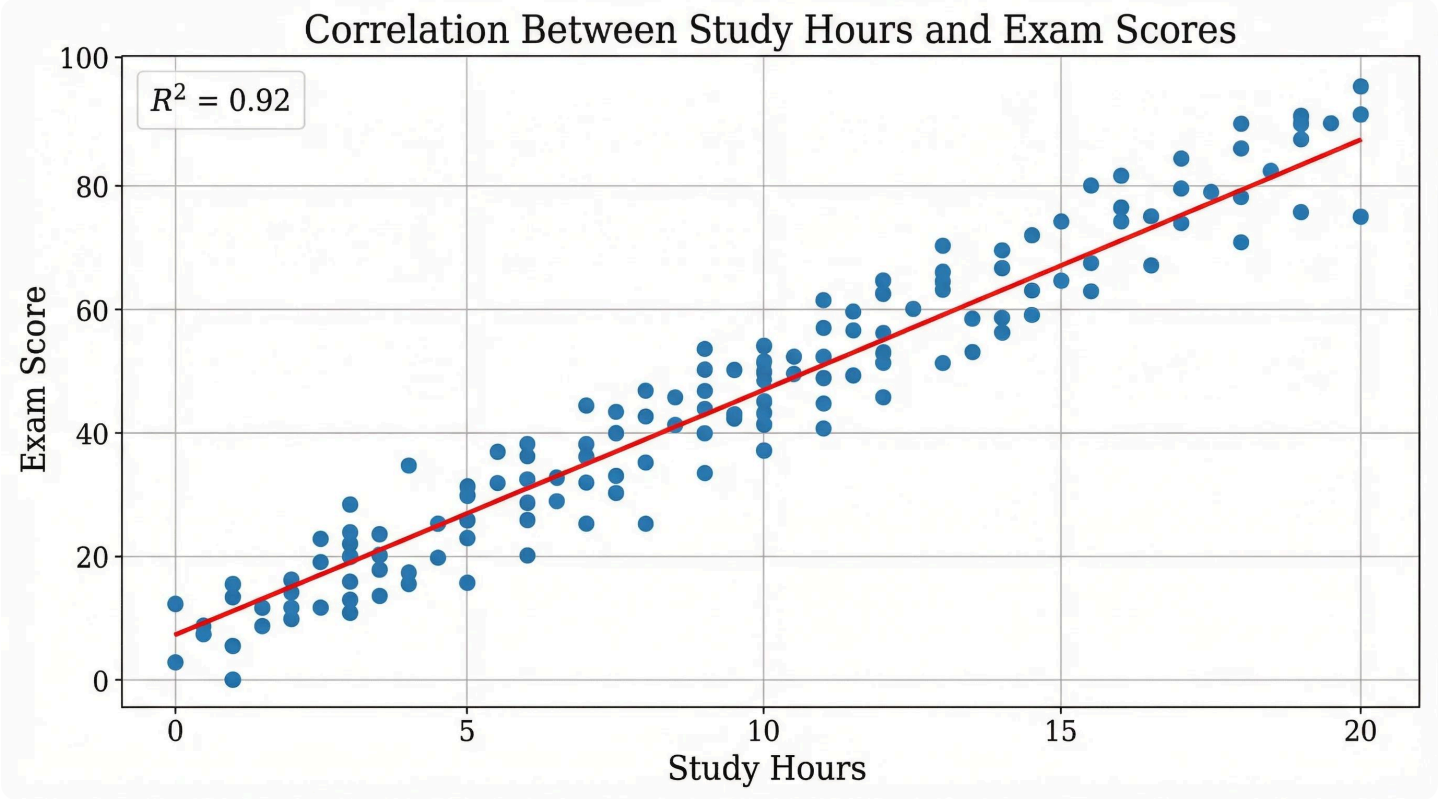
MEAN < MEDIAN < MODE

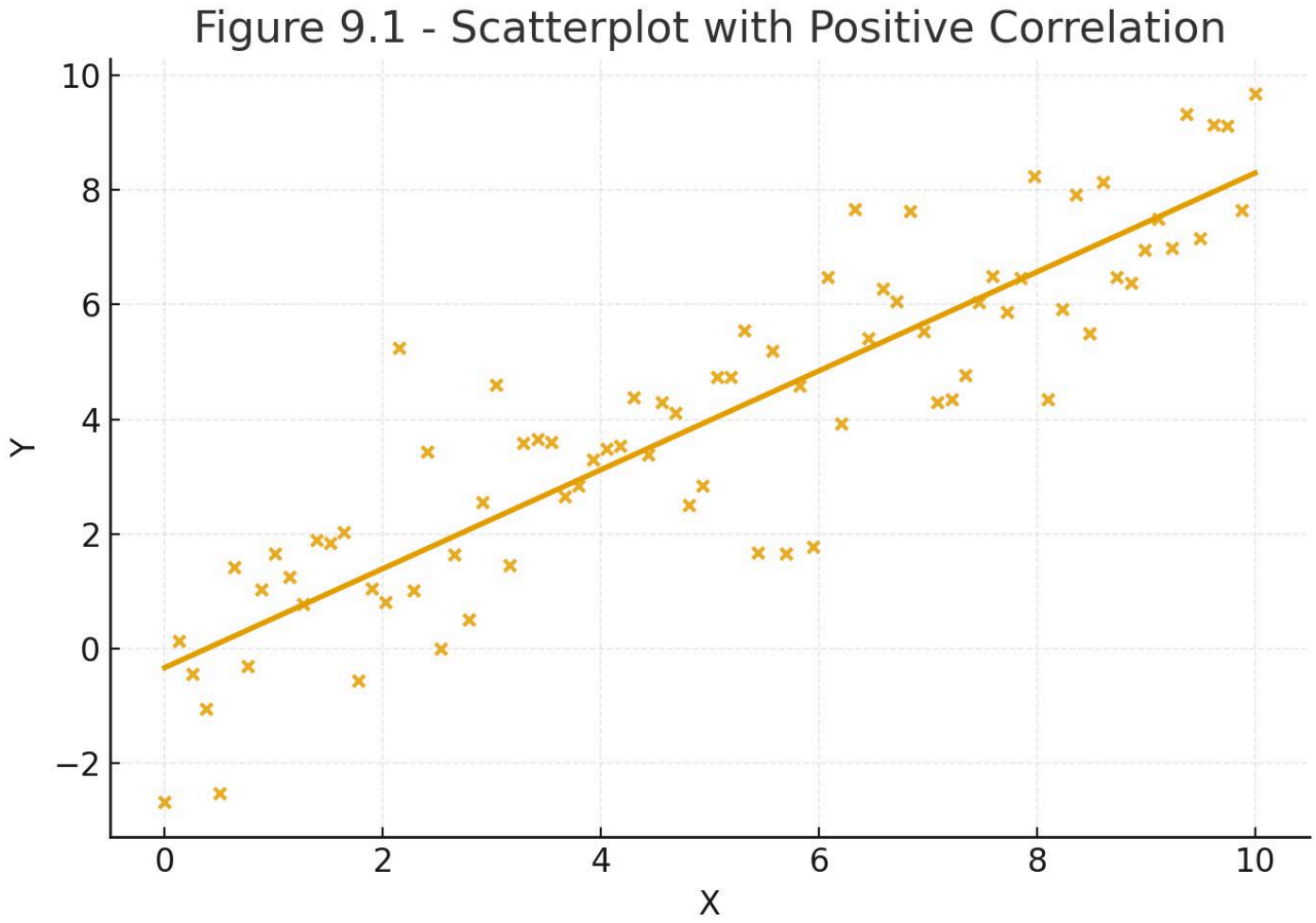



Distribution of Exam Scores
Frequency
0
10
20
30
40
50
60
0
10
20
30
40
50
60
70
80
90
100
Score


Quantiles of age
20
40
60
80
100
0
.25
.5
.75
1
Fraction of the data

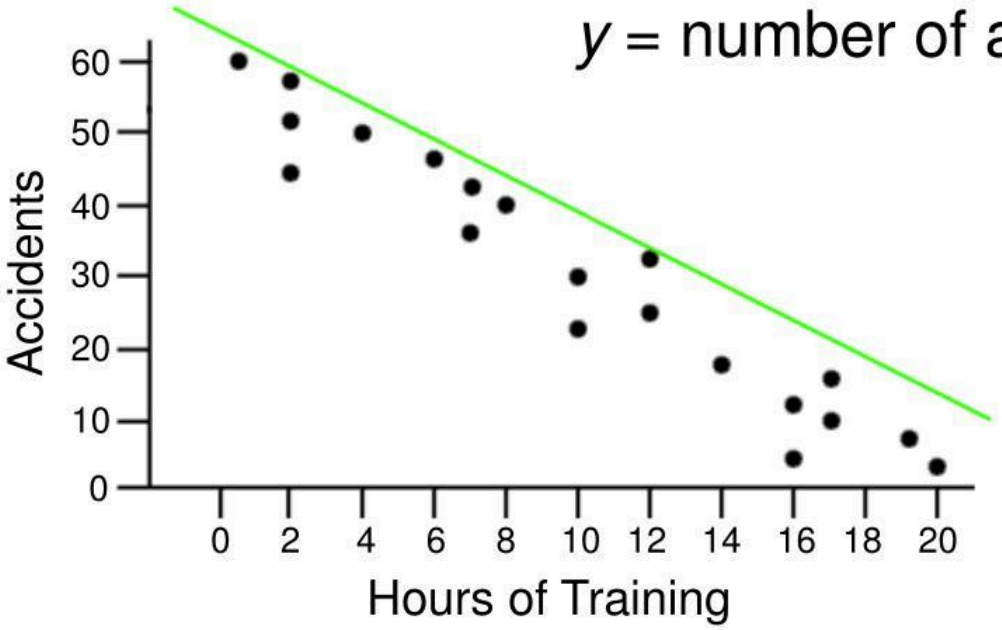

Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3


Correlation Between Study Hours and Exam Scores
R^2 = 0.92
Exam Score
Study Hours
100
80
60
40
20
0
0
5
10
15
20


Figure 9.1 - Scatterplot with Positive Correlation
10
8
6
4
2
0
−2
Y
0
2
4
6
8
10
X

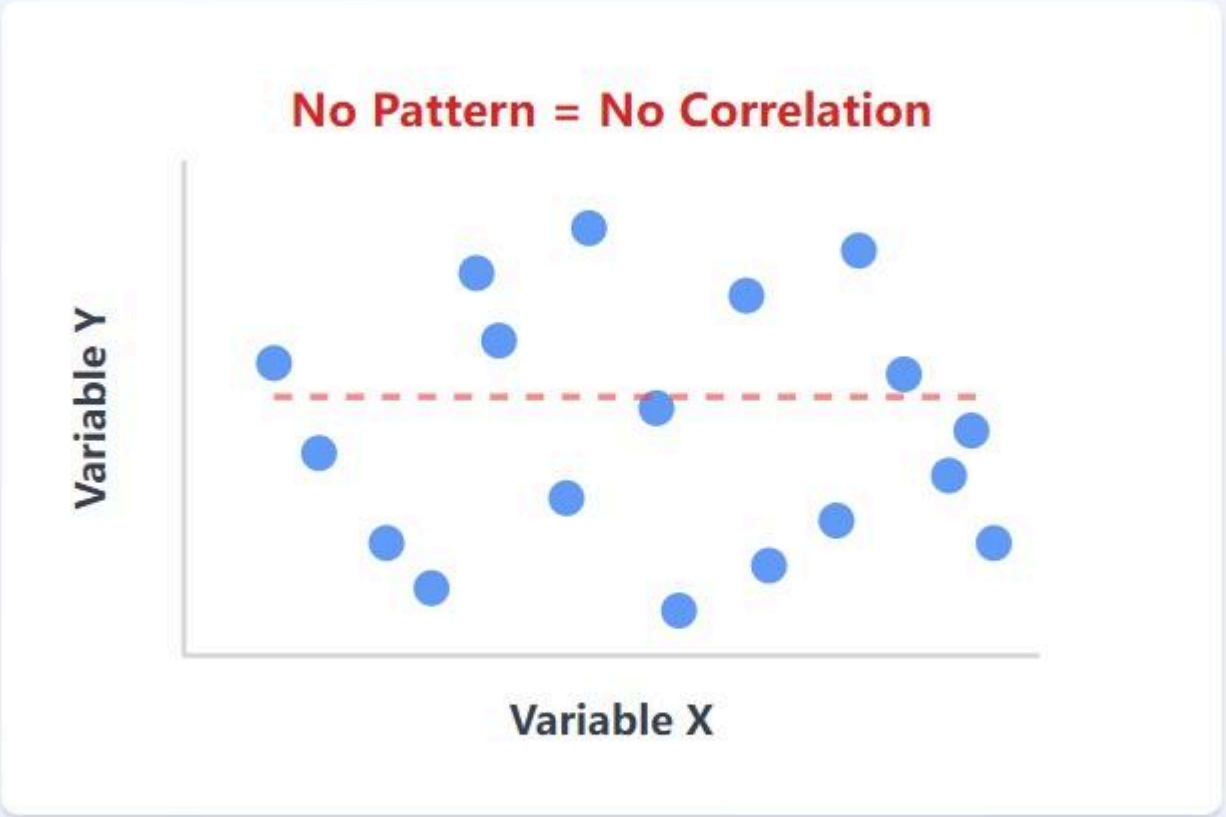
Scatter Plots and Types of Correlation

x = hours of training
y = number of accidents



Negative Correlation–as x increases, y decreases

What No Correlation Looks Like

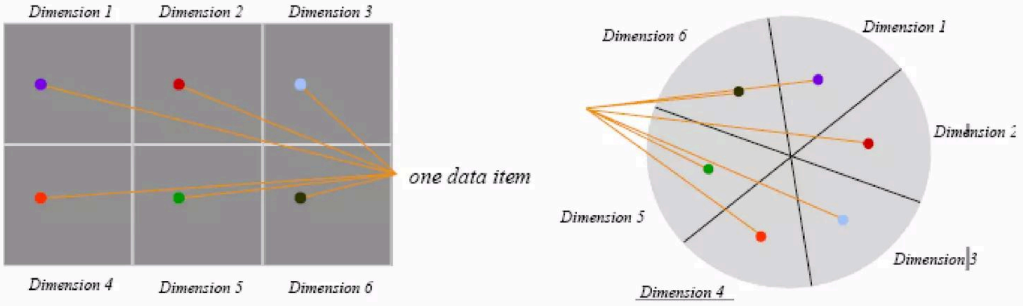


Random scatter with no clear pattern indicates that the variables are independent. Changes in X do not predict changes in Y.

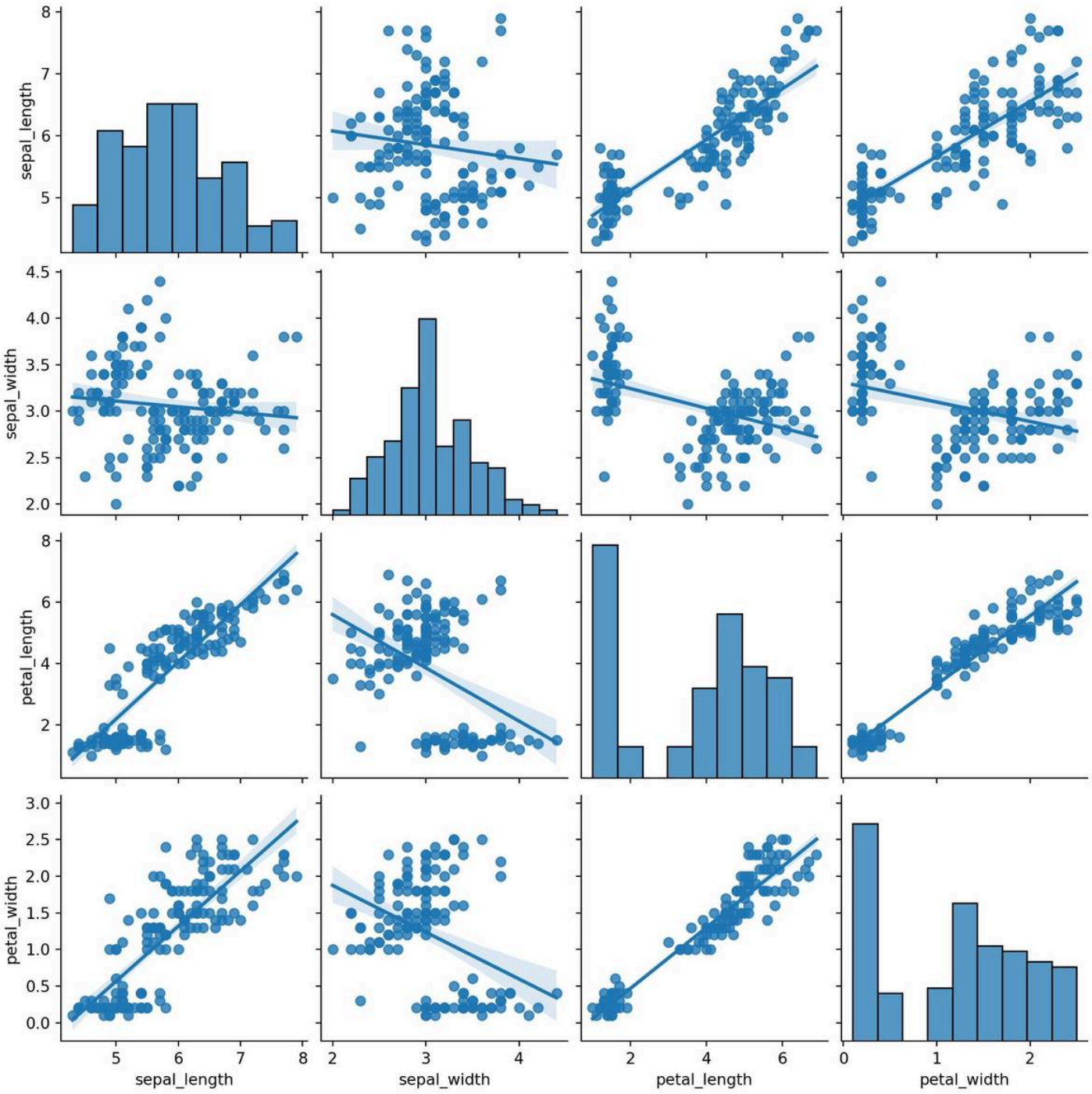
Dense Pixel Display

Approach:

- Each attribute value is represented by one colored pixel (the value ranges of the attributes are mapped to a fixed color map).
- Different attributes are presented in separate sub windows.

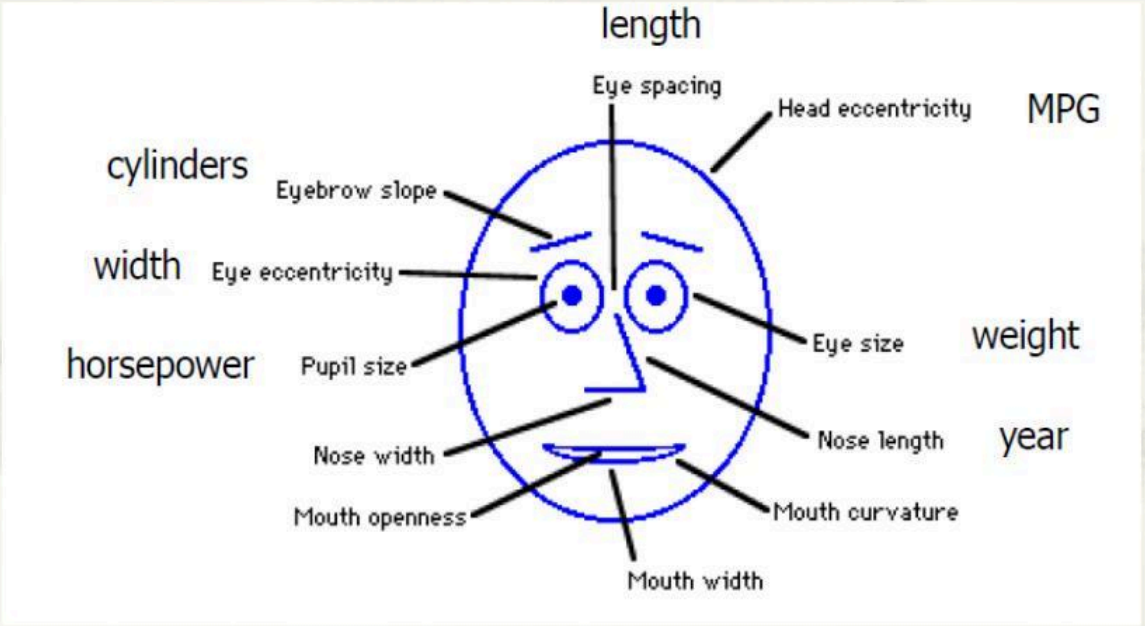


14


sepal_length
sepal_width
petal_length
petal_width
sepal_length
sepal_width
petal_length
petal_width

高维数据可视化—
切尔诺夫脸谱图(Chernoff Faces)

用人脸特征编码不同变量的值



E-commerce Sales by Category and Subcategory

